



# PROJECT REPORT ON

# SMART RESUME WRITER AND COVER LETTER BUILDER USING GenAI

AT

# Fratello Innotech Pvt. Ltd

Project Report submitted in partial fulfillment of the requirements for the award of the Degree of

# MASTER OF COMPUTER APPLICATIONS

Submitted by  
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Under the Supervision of  
**Mr. MOHAMMEDISMAIL**

## 2024 – 2026 Batch



## **CERTIFICATE**

This is to certify that the project work entitled

### **SMART RESUME WRITER AND COVER LETTER BUILDER USING GenAI**

Is a bonafide work done by

**GAMPALA VARSHITHA**  
**( 1325-24-862-207 )**

as part of the curriculum in the

DEPARTMENT OF COMPUTER SCIENCE  
AURORA'S POST GRADUATE COLLEGE (MCA)  
Ramanthapur, Hyderabad – 500013

In partial fulfillment of requirement for award of

**Master of Computer Applications**

Osmania University, Hyderabad

This work has been carried out under our guidance.

Internal Guide

Head of Department

Principal

Internal Examiner

External Examiner

**Date : 15-JUN-2026**

**TO WHOM IT MAY CONCERN**

This is to certify that Ms. **Gampala Varshitha**, student of **Aurora's Pg college (MCA)**, studying in **MCA** with HT.NO. **1325-24-862-207** has successfully completed internship at Fratello Innotech Pvt Ltd from **03-MAR-2026** to **15-JUN-2026** under the guidance of **Sindhu**.

She has worked on a project title **“Smart Resume Writer And Cover Letter Builder Using Gen AI”**. During the period of her internship program with us she has been exposed to different processes and technologies.

During her internship she has demonstrated her skills with self-motivation to learn new skills. Her performance exceeded our expectations and she was able to complete the tasks on time and was found diligent, hardworking and sincere.

We wish her all the best for her upcoming career.

**For Fratello InnoTech Pvt Ltd**



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## **DECLARATION**

I, **GAMPALA VARSHITHA**, hereby declare that the project entitled “**SMART RESUME WRITER AND COVER LETTER BUILDER USING GenAI**” has been independently carried out by me during my academic internship at **Fratello Innotech Pvt. Ltd.** This project is submitted to **Osmania University**, Hyderabad, in partial fulfillment of the requirements for the award of the degree of **Master of Computer Applications (MCA)**.

This project work is the outcome of my own efforts, research, and practical learning, and it reflects the knowledge and skills acquired throughout the duration of my postgraduate studies. The primary objective of this project is to develop an intelligent platform that leverages Generative Artificial Intelligence (GenAI) to assist users in creating professional resumes and personalized cover letters efficiently. The system aims to simplify the job application process by generating customized, industry-relevant content based on user inputs, thereby enhancing employability and career opportunities.

I further declare that the content of this dissertation is original and has not been submitted, either wholly or in part, to any other university or institution for the award of any degree, diploma, or academic recognition. All sources of information and references used have been duly acknowledged.

This project has provided me with valuable exposure to real-time software development practices and has enhanced my understanding of Artificial Intelligence, Natural Language Processing, prompt engineering, and the implementation of intelligent solutions for real-world career development and recruitment challenges.

(Signature of the student)

**GAMPALA VARSHITHA**

**(1325-24-862-207)**

## **ACKNOWLEDGEMENT**

I would like to express deep gratitude and respect to all those people behind the scene who guided, inspired in the completion of this project work.

I wish to convey my sincere thanks to Principal and Head of Department Computer Science, our project in charge **Mr. MOHAMMEDISMAIL** and project guide **Mr. RAMANJANEYULU** for giving me the required guidance during this project work.

Last but not least I am very thankful to the faculty members of my college and friends for their suggestions and help me successfully completing this project.

**GAMPALA VARSHITHA**

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# **ORGANIZATION PROFILE**

# **1.Organization Profile (Fratello Innotech Pvt. Ltd)**

Fratello Innotech Pvt. Ltd., founded in 2014 and headquartered at KPHB, Hyderabad, has established itself as a dynamic force in the development of Mobile Applications and IoT projects. With a strong focus on tackling modern-day technological challenges, we specialize in software development, with continued dedication to achieving excellence in the design, development, and maintenance of software products and solutions across diverse domains.

In today's fast-paced and competitive tech landscape, students and aspiring professionals in the Computer Science stream seek organizations that not only nurture their potential but also offer real-world exposure. At Fratello Innotech, we provide such an environment with our team of highly skilled and experienced IT professionals who specialize in studying, designing, developing, enhancing, customizing, implementing, and maintaining robust software systems.

We pride ourselves on understanding customer needs and enhancing their professional quality of life by making advanced technology accessible, efficient, and user-friendly. Our core expertise spans across:

- Software Development
- Network Simulation
- Search Engine Optimization (SEO)
- Customization
- System Integration

Our project methodology incorporates best practices for:

- Project initiation and requirement gathering
- Clear task assignments
- Dynamic scheduling
- Regular executive reporting
- Efficient problem-solving strategies

## About the Team

As a collective, our team operates with a clear vision and a deep commitment to realizing it. Backed by over 40,000 hours of cumulative experience, we excel in delivering real-time solutions in a wide array of technologies and domains, including:

- **Mobile App Development:** Android, iOS, and Flutter
- **Web Design & Development**
- **Cloud Computing**
- **Image Processing & Implementation**
- **Client-Server Technologies:** Java (J2EE), Android, Python, PHP
- **Database Systems:** Oracle, MySQL

Each team member brings a wealth of knowledge, passion, and precision to every project, ensuring our solutions are not only technically sound but also impactful and scalable.

## Our Vision

**“Turning the Impossible into Possible”**

This is more than just a motto—it is the guiding principle that drives our innovation. At Fratello Innotech, we are committed to redefining boundaries through ingenuity, dedication, and a passion for excellence. We believe every challenge presents an opportunity to learn, grow, and create solutions that shape the future.

# **ABSTRACT**

## 2. Abstract

The Smart Resume Writer and Cover Letter Builder Using GenAI is a web-based application designed to help users create professional resumes and cover letters efficiently using Generative Artificial Intelligence. In today's competitive job market, creating a well-structured resume and personalized cover letter is essential, but it can be time-consuming and challenging. This system provides an intelligent and automated solution to simplify this process.

The system consists of two main modules: Admin and Customer. The Admin manages the platform by approving customer registrations, viewing customer details, managing contacts, adding resume templates (files), and publishing notifications and job notifications. The system also includes a smart feature where job notifications are automatically sent to customers via email based on their skills and profile matching.

The Customers can register, log in, and enter their personal, academic, and professional details such as skills, projects, certifications, and work experience.

Users can generate resumes and cover letters using GenAI through two modes: Automatic, which uses stored profile data, and Manual, where users can input details and select different templates and styles. Customers can preview, download, and upload resumes, view job notifications, apply for jobs, and access templates provided by the admin.

The GenAI feature enables users to generate high-quality, well-formatted resumes and cover letters instantly, improving efficiency and enhancing job application success. Overall, the system provides a smart, user-friendly, and automated platform for resume building and career support.

The system uses artificial intelligence to analyze user-provided information such as education, skills, experience, and career goals. Based on this data, it generates well-structured, ATS-friendly resumes and personalized cover letters tailored to specific job roles. The application reduces the time and effort required for document preparation while improving quality and relevance. By leveraging GenAI technology, the system enhances users' chances of securing interviews and achieving career success.

# **SYSTEM ANALYSIS**

## 3. System Analysis

### 3.1. Existing System

In the existing system, resume and cover letter creation is mostly done manually by users using basic tools like text editors or online templates. Users need to design, format, and write content on their own, which requires time, effort, and proper knowledge of resume structure.

There is no intelligent system to automatically generate resumes or cover letters based on user data. Users often face difficulties in organizing their personal, academic, and professional details in a proper format. Additionally, selecting suitable templates and customizing resumes is a complex process for many users.

Job notifications are usually searched manually on different platforms, and there is no system that automatically matches job opportunities with user skills and sends alerts. Communication between users and administrators is also limited, and there is no centralized platform to manage resumes, templates, and notifications efficiently.

Overall, the existing system lacks automation, intelligent content generation, centralized management, and personalized job notification features, making the resume creation process time-consuming and less effective.

### Drawbacks of Existing System

The existing system for resume creation is mostly manual and time-consuming, requiring users to write, format, and edit resumes on their own. There is no AI support to automatically generate resumes or cover letters, which makes the process slower and less efficient. Many users lack proper knowledge of professional formatting and find it difficult to choose and customize templates. Additionally, there is no centralized system to manage personal details, resumes, and templates in one place. Users must manually search for jobs without any skill-based matching or automatic notifications. The system also offers limited personalization, making it hard to create customized resumes. Overall, managing and updating information becomes difficult, and the absence of instant resume generation leads to low productivity.

### 3.2 Proposed System

The proposed **Smart Resume Writer and Cover Letter Builder Using GenAI** is a web-based application designed to automate and simplify the process of creating professional resumes and cover letters. This system uses Generative Artificial Intelligence to generate high-quality documents based on user-provided data.

In this system, the **Admin** manages the entire platform by approving customer registrations, viewing customer details, managing contact messages, adding resume templates (files), and publishing notifications and job notifications. The system includes a smart feature where job notifications are automatically sent to customers via email based on their skills and profile matching. The admin can also manage templates and ensure proper system functionality.

The **Customers** can register and log in after admin approval. They can enter their personal, academic, and professional details such as skills, projects, certifications, and work experience. The system allows users to generate resumes and cover letters using GenAI with two options: **Automatic mode**, which uses stored profile data, and **Manual mode**, where users can input details and select different templates and styles.

Customers can preview, download, and upload resumes, view job notifications, apply for jobs, and access templates added by the admin. The system provides a user-friendly interface for managing all resume-related data and ensures secure account management

## **Advantages of Proposed System**

**AI-Based Resume Generation** – Automatically generates professional resumes and cover letters.

**Saves Time and Effort** – Reduces manual work in writing and formatting documents.

**Automatic and Manual Modes** – Users can generate resumes using stored data or by entering details manually.

**Professional Templates** – Provides multiple resume templates and styles for better presentation.

**Centralized Data Management** – All user details and resumes are stored in one system.

**Skill-Based Job Notifications** – Sends job alerts automatically based on user skills.

**Email Notification System** – Job notifications are sent directly to users via email.

**User-Friendly Interface** – Easy to use for all types of users.

**Profile Management** – Users can manage and update personal and professional details.

**Resume Preview and Download** – Users can preview and download resumes easily.

**Upload and Manage Resumes** – Users can upload and view their resumes anytime.

**Improved Resume Quality** – AI helps in generating well-structured and professional content.

**Secure System** – Provides login authentication and password management.

**Better Job Opportunities** – Helps users find relevant jobs easily.

**Scalable System** – Can be enhanced with more advanced AI features in the future.

### 3.3.Module Description

#### Admin Modules

The Admin Module is responsible for managing the entire system. The admin can log in securely and approve customer registrations before allowing access to the system. The admin can view customer details, manage contact messages, add and view resume templates (files), and publish notifications and job notifications. The admin also ensures proper system functionality and can change the password and log out securely

#### Customers Modules

The Customer Module allows users to register and log in after admin approval. Customers can manage their profiles by adding personal, academic, and professional details such as skills, projects, certifications, and work experience. Users can view notifications, job notifications, and templates provided by the admin. They can also upload, view, and download resumes, apply for job notifications, and manage their accounts securely

#### Scope and Objective of the Project

The main objective of the **Smart Resume Writer and Cover Letter Builder Using GenAI** is to develop a web-based system that helps users create professional resumes and cover letters using Generative Artificial Intelligence. The system aims to reduce manual effort and provide a quick and efficient way to generate high-quality documents.

Another objective is to provide a centralized platform where users can store and manage their personal, academic, and professional details. The system also focuses on improving job opportunities by sending skill-based job notifications to users through email. Additionally, the admin can manage users, templates, notifications, and job postings to ensure smooth system operation

The scope of the **Smart Resume Writer and Cover Letter Builder Using GenAI** includes user registration and login with admin approval, resume and cover letter generation using AI, and management of user profiles. Customers can enter their details, generate resumes in automatic or manual mode, select templates, preview, download, and upload resumes.

The system also includes features such as template management, notification services, and job notifications with automatic email alerts based on skill matching. The admin manages all activities, including approving users, adding templates, and posting job opportunities. In the future, the system can be enhanced with advanced and integration with job portals to improve functionality and user experience

### 3.4 .System Requirements and Specifications

#### Software Requirements

- Operating System : Windows
- User Interface : HTML, CSS , JS and Bootstrap
- Programming Language : python
- Web Applications : Django
- IDE/Workbench : Pycharm
- Database : MYSQL, SQL, SQLYOG
- Server Deployment : Apache Tomcat

#### Hardware Requirements

- Processor : Intel I3 Processor
- RAM : 4 GB
- Monitor : 15 inch color monitor or LED
- Hard disk : 256 GB
- Key board : Standard 102 keys
- Mouse : Optical

#### Functional Requirements:

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So it's essential to make them clear both for the development team and the stakeholders. Generally, functional requirements describe system behavior under specific conditions.

HOME

ABOUT

CONTACT

ADMIN

➤ Admin Functions:

- Login
- Homepage
- View Customer
- View Contacts

- Add Files
- View Files
- Add Notification
- View Notification
- Add Job Notification
- View Job Notifications
- Change Password
- Logout

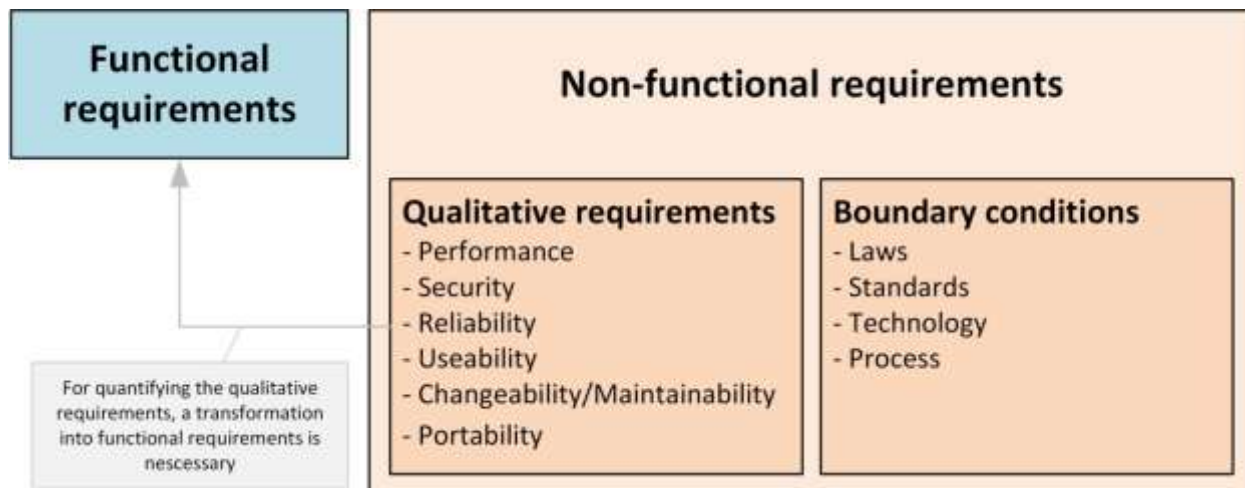
## **CUSTOMER**

### ➤ Customer Functions:

- Register
- Login
- Homepage
- profile
- Upload Resume
- View Resume
- View Notification
- View Job Notifications
- View Files
- View Personal Details
- View Carrier Summary
- View Academic Qualifications
- View Hobbies
- View Project Details
- View Technical Skills
- Language Known
- Work Experience
- View Achievements
- View Certifications
- View Declaration
- View Resume
- Create Resume
- Change Password
- Logout

## Non Functional Requirements:

Non functional testing is a type of software testing that verifies non functional aspects of the product, such as performance, stability, and usability. Whereas functional testing verifies whether or not the product does what it is supposed to, non functional testing verifies how well the product performs.



## The system should shows below non functional requirements

1. High Security
2. Multi browser and operating system compatibility
3. Mobile responsiveness
4. High quality and performance
5. Memory optimized and User Friendliness
6. Easy maintenance and scalable capability
7. It's should support good accessibility and usability
8. It should follow regulatory and compliance policies

### **3.5.Feasibility study**

The possibility of the undertaking is broke down during this stage and strategic plan is advanced with an exceptionally broad arrangement for the task and a few quotes. During project examination the likelihood investigation of the proposed project is to be completed. this is often to ensure that the proposed project isn't a weight to the organization. For practicality examination, some comprehension of the many necessities for the project is prime.

After the approval of the request to the organization and project guide, with an investigation being considered, the project request must be examined to determine precisely what the system requires. Not all request projects are desirable or feasible. Some organization receives so many project requests from client users that only few of them are pursued. However, those projects that are both feasible and desirable should be put into schedule. After a project request is approved, it cost, priority, completion time and personnel requirement is estimated and used to determine where to add it to any project list. Truly speaking, the approval of those above factors, development works can be launched

An important outcome of preliminary investigation is the determination that the system request is feasible. This is possible only if it is feasible within limited resource and time. The different feasibilities that have to be analyzed are.

#### **ECONOMICAL FEASIBILITY**

This examination is completed to see the financial effect that the framework will wear the association. The measure of store that the organization can fill the innovative work of the framework is constrained. The consumptions must be advocated. Accordingly, the created framework also inside the budget and this was accomplished in light of the very fact that an outsized portion of the advancements utilized are uninhibitedly accessible. Just the modified items must be bought. Economic Feasibility or Cost-benefit is an assessment of the economic justification for a computer based project. As hardware was installed from the beginning & for lots of purposes thus the cost on project of hardware is low. And also in this project work all software is free and open source. It can be developed using our personal machines. So its economically its feasible to complete the project.

## **TECHNICAL FEASIBILITY**

According to Roger S. Pressman, Technical Feasibility is the assessment of the technical resources of the organization. The system is developed for platform Independent environment. The technical feasibility has been carried out. The system is technically feasible for development and can be developed with the existing facility.

This investigation is completed to see the technical practicality, that is, the specialized prerequisites of the framework. Any framework created must not have a popularity on the accessible specialized assets. this may prompt high requests on the accessible specialized assets. this may prompt high requests being assail the customer. The created framework must have a humble prerequisite, as just insignificant or invalid changes are required for actualizing this framework.

## **SOCIAL FEASIBILITY**

The a part of study is to see the degree of acknowledgment of the framework by the client. This incorporates the way toward preparing the client to utilize the framework effectively. The client must not feel undermined by the framework, rather should acknowledge it as a requirement . The degree of acknowledgment by the clients exclusively relies upon the techniques that are utilized to show the client about the framework and to form him familiar with it. His degree of certainty must be raised with the goal that he's likewise able to make some helpful analysis, which is invited, as he's the last client of the framework.

## **OPERATIONAL FEASIBILITY**

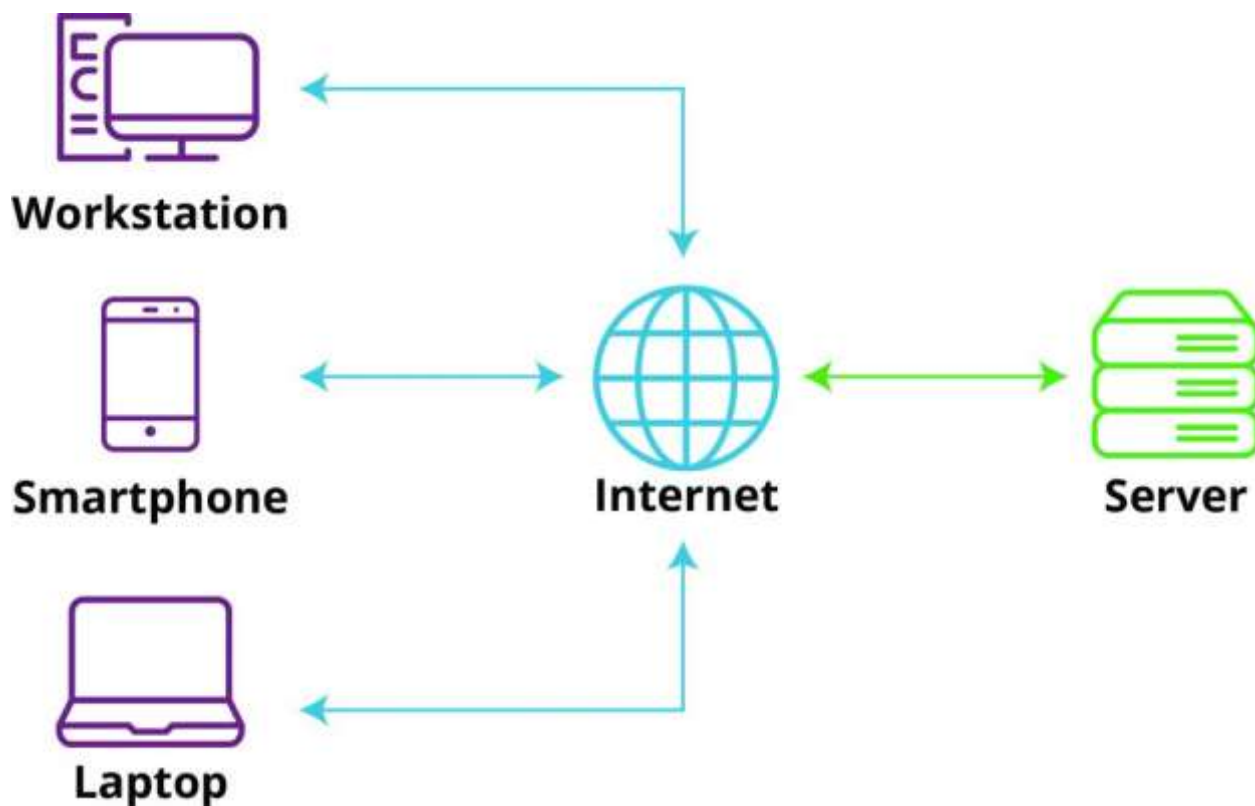
This region of attainability ensures that the undertaking fulfills all the prerequisites that were talks about during introductory necessity examination stage. This undertaking met all the necessities proposed before. It ensured that the modules examined during the proposing of the framework were actualized with stable precision and with no mistakes.

Operational Feasibility deals with the study of prospects of the system to be developed. This system operationally eliminates all the tensions of the Admin and helps him in effectively tracking the project progress. This kind of automation will surely reduce the time and energy, which previously consumed in manual work. Based on the study, the system is proved to be operationally feasible.

## Client Server Architecture

Client server architecture, architecture of a computer network in which many clients (remote processors) request and receive service from a centralized server (host computer). Client computers provide an interface to allow a computer user to request services of the server and to display the results the server returns.

The Client-server model is a distributed application structure that partitions tasks or workloads between the providers of a resource or service, called servers, and service requesters called clients. In the client-server architecture, when the client computer sends a request for data to the server through the internet, the server accepts the requested process and delivers the data packets requested back to the client. Clients do not share any of their resources. Examples of the Client-Server Model are Email, World Wide Web, etc.



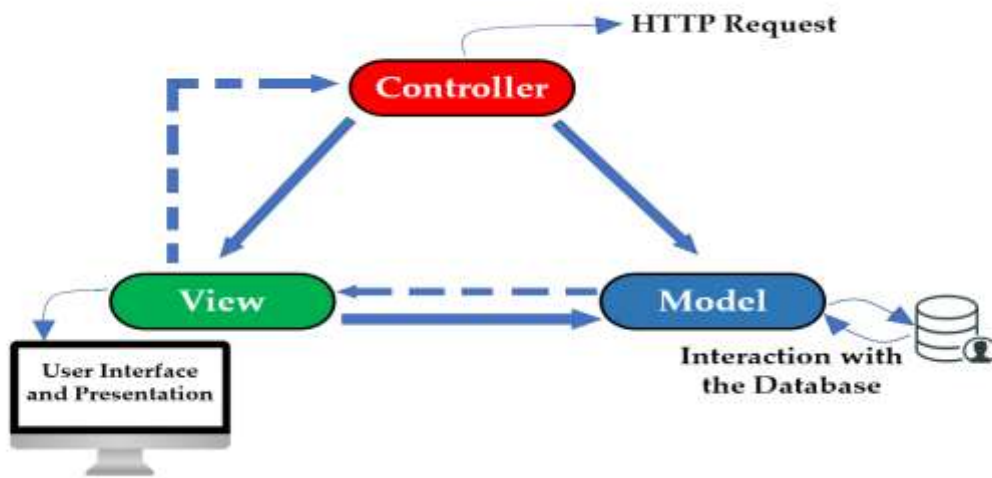
## Web Application MVC Architecture

The Model-View-Controller (MVC) is a well-known design pattern in the web development field. It is a way to organize our code. It specifies that a program or application shall consist of data model, presentation information and control information. The MVC pattern needs all these components to be separated as different objects.

The MVC pattern architecture consists of three layers:

- **Model:** It represents the business layer of application. It is an object to carry the data that can also contain the logic to update controller if data is changed.
- **View:** It represents the presentation layer of application. It is used to visualize the data that the model contains.
- **Controller:** It works on both the model and view. It is used to manage the flow of application, i.e. data flow in the model object and to update the view whenever data is changed.

### **Model-View-Controller Architecture Block Diagram**



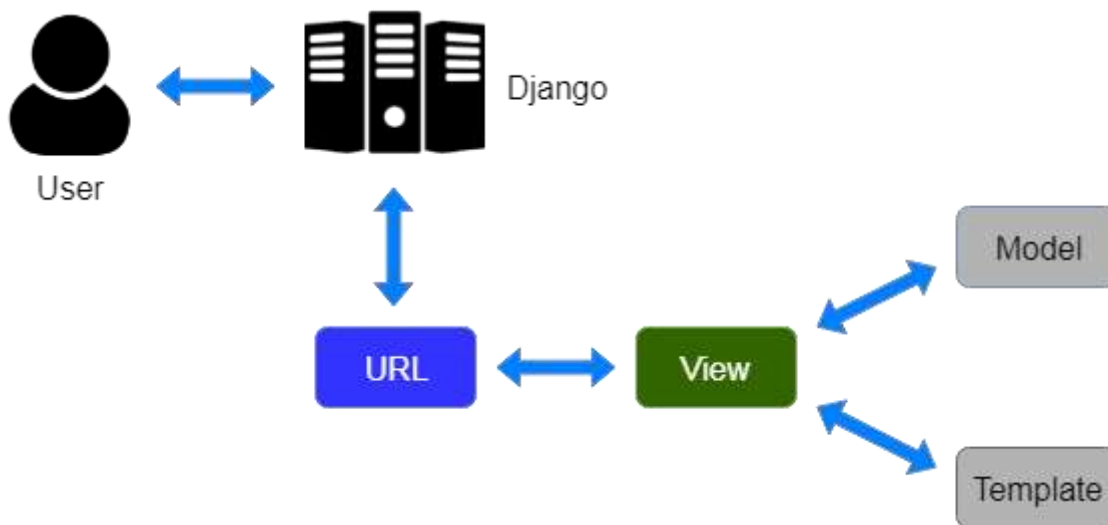
### **Web Application MVT Architecture**

The MVT (Model View Template) is a software design pattern. It is a collection of three important components Model View and Template. The Model helps to handle database. It is a data access layer which handles the data.

The Template is a presentation layer which handles User Interface part completely. The View is used to execute the business logic and interact with a model to carry data and renders a template.

The model is going to act as the interface of your data. It is responsible for maintaining data. It is the logical data structure behind the entire application and is represented by a database (generally relational databases such as MySQL, Postgres).

The View is the user interface — what you see in your browser when you render a website. It is represented by HTML/CSS/Javascript and Jinja files.



## Glossary:

### Accessibility

Accessibility refers to the usability of a website or web application for all users, including those with disabilities.

### Agile

Agile is a commonly used web development approach that focuses on flexibility and adaptability, helping teams quickly respond to changes in strategy and requirements.

### Algorithm

As a web development term, an algorithm is a set of instructions used to automate tasks.

Algorithms can include searching for and displaying content, sorting data and scanning for bugs or mistakes.

### Application Program Interfaces (APIs)

APIs are a set of definitions and protocols for developers to rely on when building and integrating web applications.

### Backend Development

Backend development is one of the most central web development terms you will come across during any website development project.

## **Bandwidth**

Bandwidth refers to the rate that data is transferred between a web server and a user's computer when loading any aspect of a website.

## **Bootstrap**

Bootstrap is one of the most popular frontend web development frameworks. Its library of design templates is built via HTML, CSS and JavaScript to create responsive, mobile-first designs with interactive elements.

## **Browser**

A browser is an application that allows both users and developers to view webpages on the internet.

It is an important aspect of any web development glossary, as browsers not only retrieve and display data on the user's device but also provide features and tools that can be utilized in the web development process.

## **Cache**

Cache is a form of web storage used in website development that helps improve the loading speed of a webpage and its elements.

## **Content Delivery Network (CDN)**

Used to improve speed, reliability and performance of web pages by caching features such as images, videos and files, a CDN is a geographically distributed set of servers.

## **Coding**

Coding is the process of writing computer code to define and instruct the layout, functionality, design, behavior and interactions on a website.

There are two distinct forms of coding:

**Frontend coding:** Using HTML and CSS, a developer will tell a browser how the main functionalities of a webpage are to be structured and designed.

**Backend coding:** Using programming languages such as PHP and .Net, developers will instruct specific features of a website or web application. These features can include security, communication and notification aspects such as log-ins and chatbots.

Elsewhere, Script coding allows for the more complex backend-coded features to take on frontend coding aesthetics and usability.

While some web development projects use themes and plugins that rely on pre-written code, other expert developers will introduce new code to create an individualized experience.

## **Cookies**

You've most likely come across the term while browsing the internet and have been forced to either accept or decline cookies after landing on a website.

## **Database**

A database is a collection of structured information organized in tables, that can be accessed by users or applications through queries.

Databases are used to store user data, such as accounts and passwords, product information, payment details and statistics. Your web development project will revolve around a database, such as MySQL or PostgreSQL.

## **Debugging**

While coding acts as the central player in bringing a website to life, it also plays a role in causing issues or bugs down the line.

Debugging is the process of locating and correcting these coding problems to ensure a site continues to run as intended.

## **Deployment**

Deployment is the process of collecting all code, databases, assets and other parts of your project and launching them to an appropriate hosting environment to be accessed by the intended users.

## **Framework**

Framework sites such as Bootstrap are popular tools, but developers can also create bespoke frameworks from scratch for their own projects by writing the HTML, CSS and JavaScript themselves.

As pre-built templates containing many helpful components and tools, developers can use frameworks to quickly create partly homogenized webpages and sites.

## **Frontend Development**

Frontend development focuses on the design and functionality of the user interface (UI) by using coding languages such as HTML, CSS and JavaScript.

## **Full Stack Development**

Full stack development is an amalgamation of all things necessary to build and maintain a website under the roof of one project.

From coding, debugging and deployment, to frontend and backend development, full stack development represents the entirety of components and tools needed to create a functioning site.

## **Git**

[Git](#) is a version control system that allows developers to keep track of their codebase, monitor changes and collaborate on projects with other developers.

## **GitHub**

[GitHub](#) is a cloud-based version control platform that provides developers with a secure and collaborative environment to store their codebase, track changes, and partner on projects with other developers.

## **Hosting**

Hosting, in relation to web development terminology, is the provision of a server or network infrastructure to store and manage webpages.

This is achieved by providing hardware, software and networks on a single platform to make websites accessible on the internet.

## **Hypertext Markup Language (HTML)**

HTML is a go-to markup language for webpage creation and structuring. If you're exploring a website, chances are that it's composed to some extent of HTML code.

Developers use HTML to tag text to achieve font, color, graphics, and hyperlink effects, as well as to implement various interactive features and elements.

## **Hypertext Transfer Protocol (HTTP)**

HTTP is the set of rules or protocols used to transfer information, such as text, images, sound and video, over the internet.

It defines how messages are formatted and transmitted, as well as how web servers and browsers should respond to various commands.

The process works by a browser sending and retrieving requests between these hypertext links and via an Internet Protocol (IP) address.

## **Integration**

Web development integration is the process of connecting different components, systems and services to enable them to work together.

## **JavaScript**

Sitting between frontend and backend coding, JavaScript is a programming language that allows developers to add interactive elements such as animations, menus and forms to a website.

## **Libraries**

Libraries for web developers, built upon the same premises as the library you will find in your hometown, contain code that has been tested and can be used to add certain functionalities to a web page or application.

Instead of collections of dusty old books, developers can find countless collections of pre-written code in these libraries, waiting to be implemented into their individual projects

## **Maintenance**

Maintenance is the process of continually ensuring a website is running smoothly and securely through fixes and updates.

Whether updating content, fixing bugs or improving features, regular maintenance plays an important role in building online credibility and trust and offering a superior user experience to your website visitors.

## **Minification**

Minification is the process of reducing the size and complexity of a website's code to improve its performance.

## **Minimum Viable Product (MVP)**

An MVP acts as a prototype for a website, with just enough features included to satisfy early users while also leaving space for feedback and further development.

## **MySQL**

[MySQL](#) is a database management system used to store and manage large amounts of data, allowing developers to quickly and easily create complex databases for their websites.

## **Operating System**

An operating system is a software program that enables communication between hardware and applications.

In web development, an operating system is responsible for running user programs, ensuring they don't interfere with one another or cause system crashes.

## **Plugins**

Plugins are software components added to provide extra functionalities to a website, which can enhance both [UX and UI](#).

Used by both designers and developers, they create the possibility of customizing a website without the need for uniquely written code.

## **Programming Language**

A programming language is a set of instructions used to write computer code for website development.

It is the foundation of how a website is created and consists of syntax, data structures, and control flow that enable developers to create complex user interfaces, databases, and website features.

## **Quality Assurance**

Quality assurance is the process of performing pre-launch checks on a website, including testing, debugging and verifying a website's code to make sure it performs reliably, security and allows for scalability.

## **Redirects**

Redirects are a mechanism used in website development to forward visitors from one URL to another.

When a visitor attempts to access a certain web page, the server will instead send them to a different page based on the rules set up in the redirect configuration.

## **Server**

The home to your hosting platform, a server stores and manages all of your website's files, applications and databases. It interacts with browsers to enable visibility on the internet, resulting in users being able to access the website.

## **Structured Query Language (SQL)**

SQL is a programming language that interacts specifically with databases, allowing developers to edit data quickly and effectively within the database.

## **Secure Sockets Layer (SSL)**

SSL is a security protocol used in website development to establish an encrypted connection, preventing third parties from intercepting or tampering with shared information.

## **Text Editor**

A text editor is a type of software used in website development to create and edit HTML, CSS, JavaScript and other web-based coding languages.

It allows developers to quickly write code, perform syntax highlighting and auto-complete functions, add debugging tools and preview what the code will look like on the website.

## **User Interface (UI)**

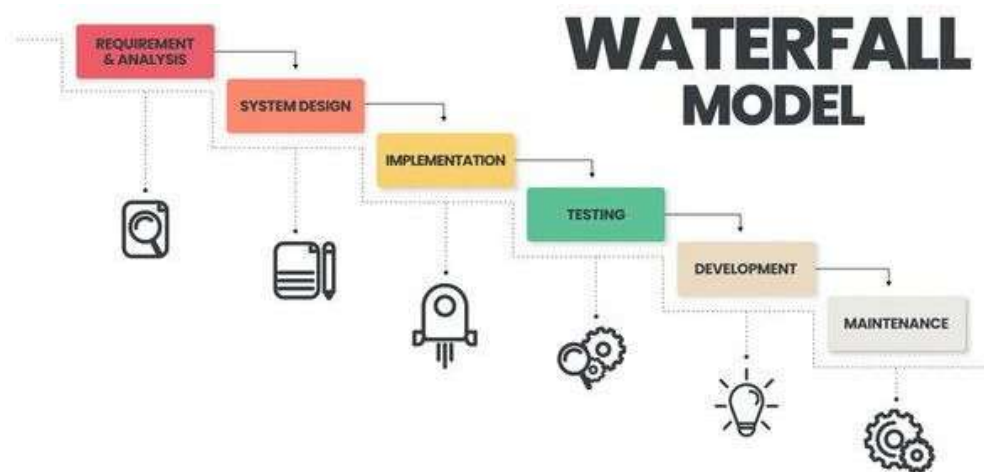
As one of the most common web development terms you will come across, UI refers to all of the elements that a user can see on a website.

## **Process Models**

### **1. Waterfall Model:**

Waterfall model indicates that all the SDLC steps will be going on step by step. In this model once one stage completed then only it will start the another process. This will be implemented when the requirements well known and clearly defined.

In this model we can implement all the steps one by one. First we will do requirements gathering i.e in this step business analyst will collect the all functional requirements from the client and will discuss with the manager and development team. Second step is analysis where people can analyses the requirements so that they will do risk basement, risk mitigation, risk controlling steps. And also they will do the cost-time analysis to estimate the cost. They will also do the detail feasible study over the requirements of project.

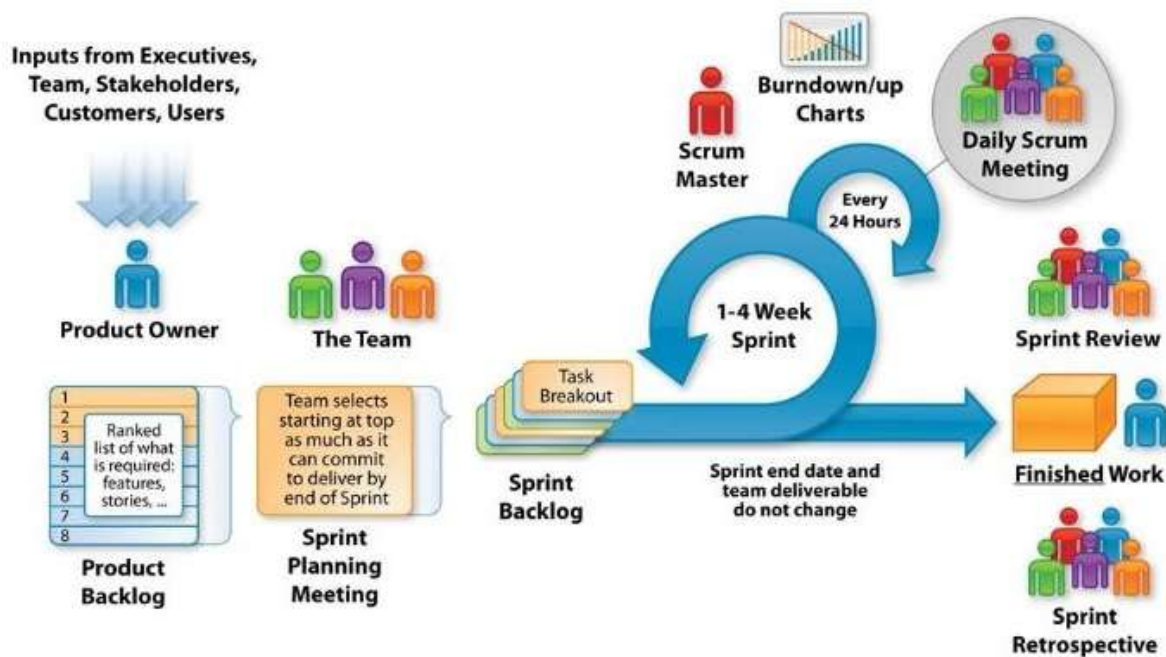


In third phase called design page generally people will design the project i.e architect will prepare the UML diagrams and prototypes for project. And then developers will start programming or coding based on design . Finally testing team will test the content to identify and fix the bugs. After successful user acceptance testing application will be deployed and maintained in the client machine. The waterfall model was selected as the SDLC model due to the following reasons:

- When the requirements are very clear and not changed.
- Available technology is enough to implement.
- Project is easy and simple to understand to finish.
- And if we don't have any confusion in requirements.
- We can easily manage the review process and task management.
- Every stage defined clearly and with timelines

## 2. Agile Scrum Model:

Agile scrum methodology is the combination of the agile philosophy and the scrum framework. Agile means “incremental, allowing teams to develop projects in small increments. Scrum is one of the many types of agile methodology, known for breaking projects down into sizable chunks called “sprints.” Agile scrum methodology is good for businesses that need to finish specific projects quickly.



Agile scrum methodology is a project management system that relies on incremental development. Each iteration consists of two- to four-week sprints, where the goal of each sprint is to build the most important features first and come out with a potentially deliverable product. More features are built into the product in subsequent sprints and are adjusted based on stakeholder and customer feedback between sprints. Whereas other project management methods emphasize building an entire product in one operation from start to finish, agile scrum methodology focuses on delivering several iterations of a product to provide stakeholders with the highest business value in the least amount of time.

Agile scrum methodology has several benefits. First, it encourages products to be built faster, since each set of goals must be completed within each sprint's time frame. It also requires frequent planning and goal setting, which helps the scrum team focus on the current sprint's objectives and increase productivity. The greatest benefit of agile scrum methodology is its flexibility. With the sprint-based model, the scrum team typically receives feedback from stakeholders after each sprint. If there are any problems or changes, the scrum team can easily and quickly adjust product goals during future sprints to provide more valuable iterations. This way, stakeholders are happier because they get exactly what they want after being involved every step of the way.

# **SYSTEM DESIGN**

## **4.System Design**

### **INPUT DESIGN**

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

### **OBJECTIVES**

- 1.Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.
- 2.It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.
- 3.When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

### **OUTPUT DESIGN**

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information

to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.
2. Select methods for presenting information.
3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the
- Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

## **4.1 UML diagram**

### **System Design:**

This section consists of the UML diagrams related to the modules developed such as Activity diagram, Sequence diagram and Use case diagram.

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

## GOALS:

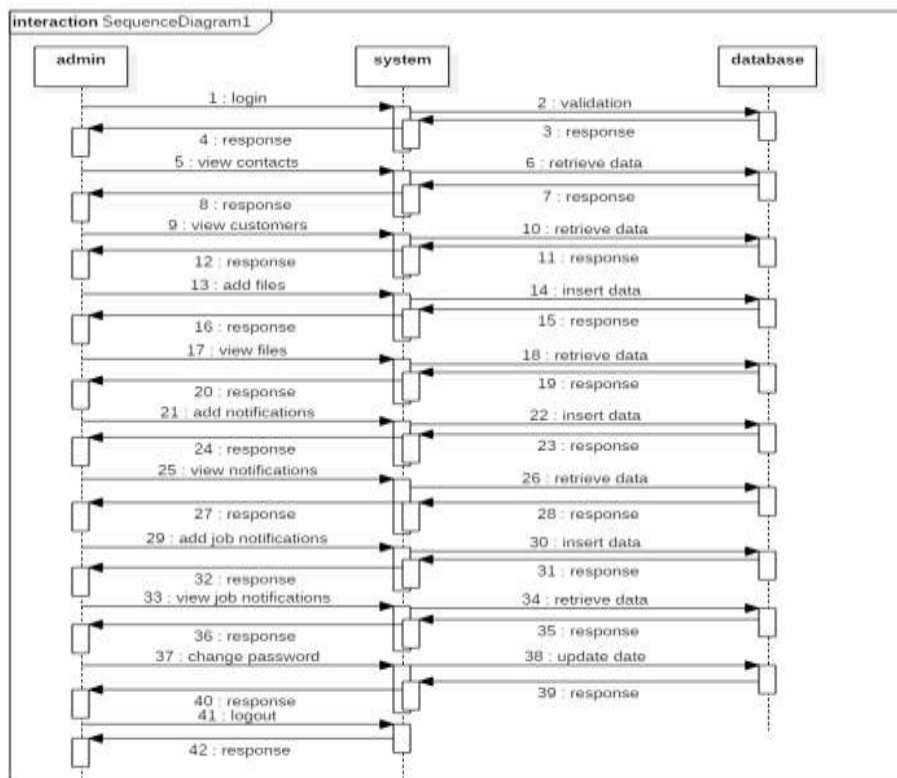
The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.

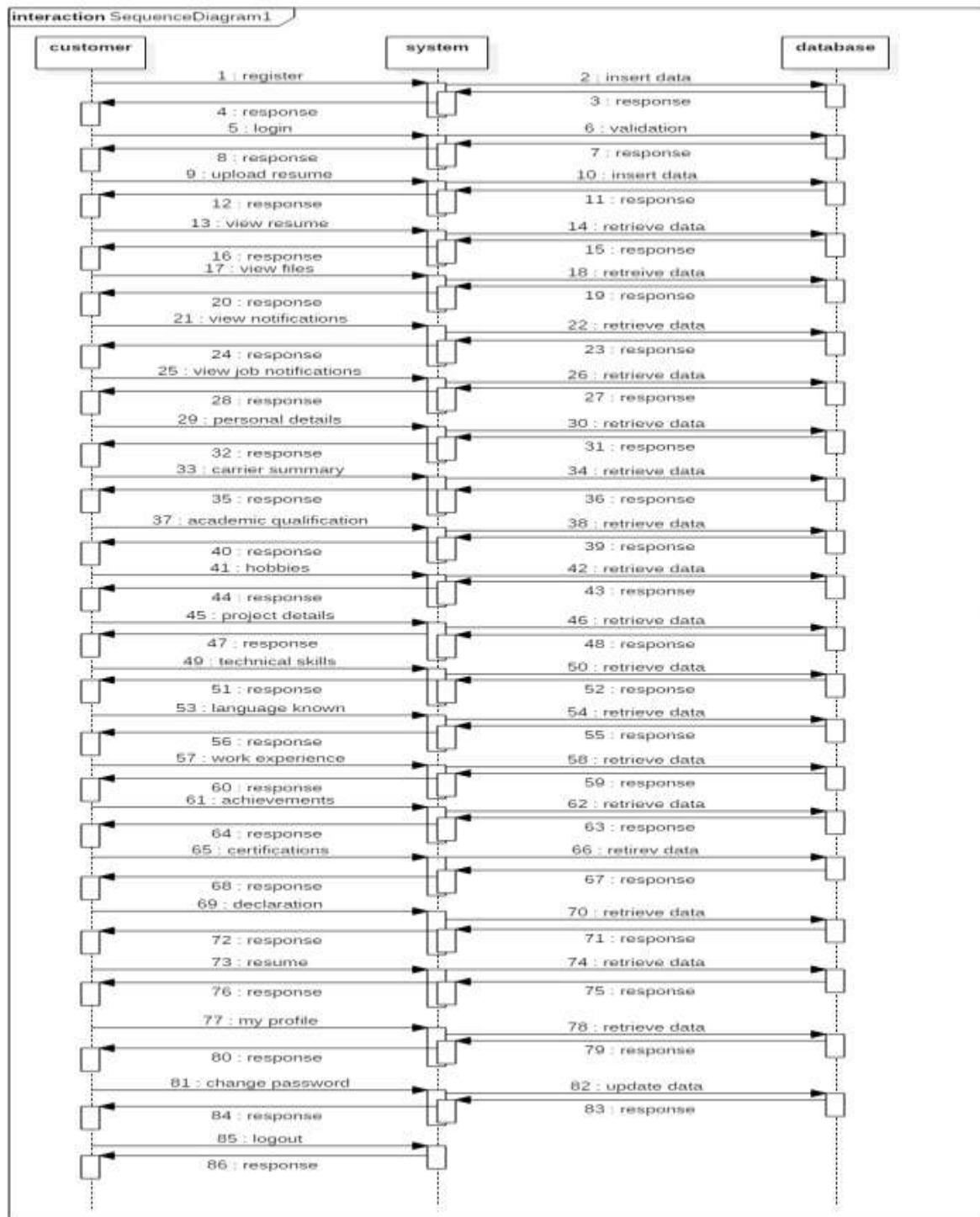
## Sequence Diagram

A grouping outline in UML is a sort of association chart that shows how procedures work with each other and in what request. “It is a build of a Message Sequence Chart. Succession outlines are now and again called occasion charts, occasion situations, and timing graphs”.

### Admin-Sequence Diagram:



## Customer Sequence Diagram:

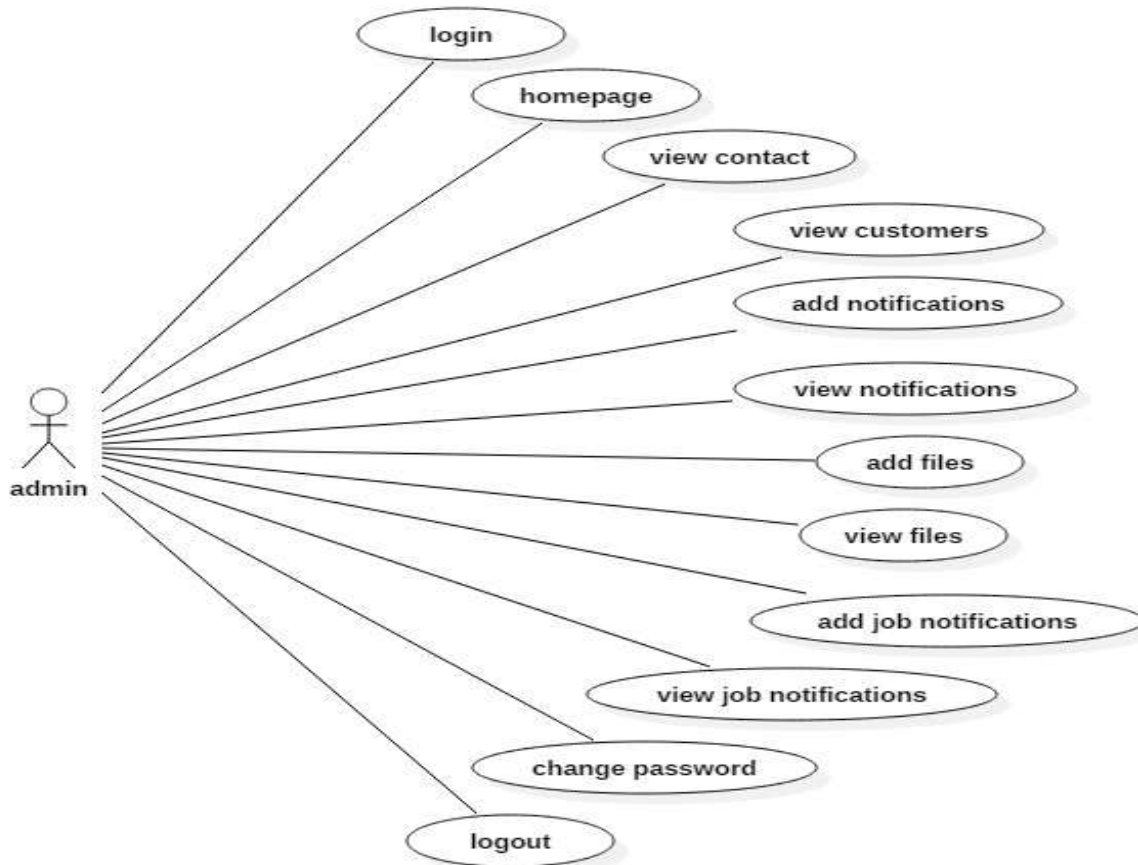


## Use Case Diagram

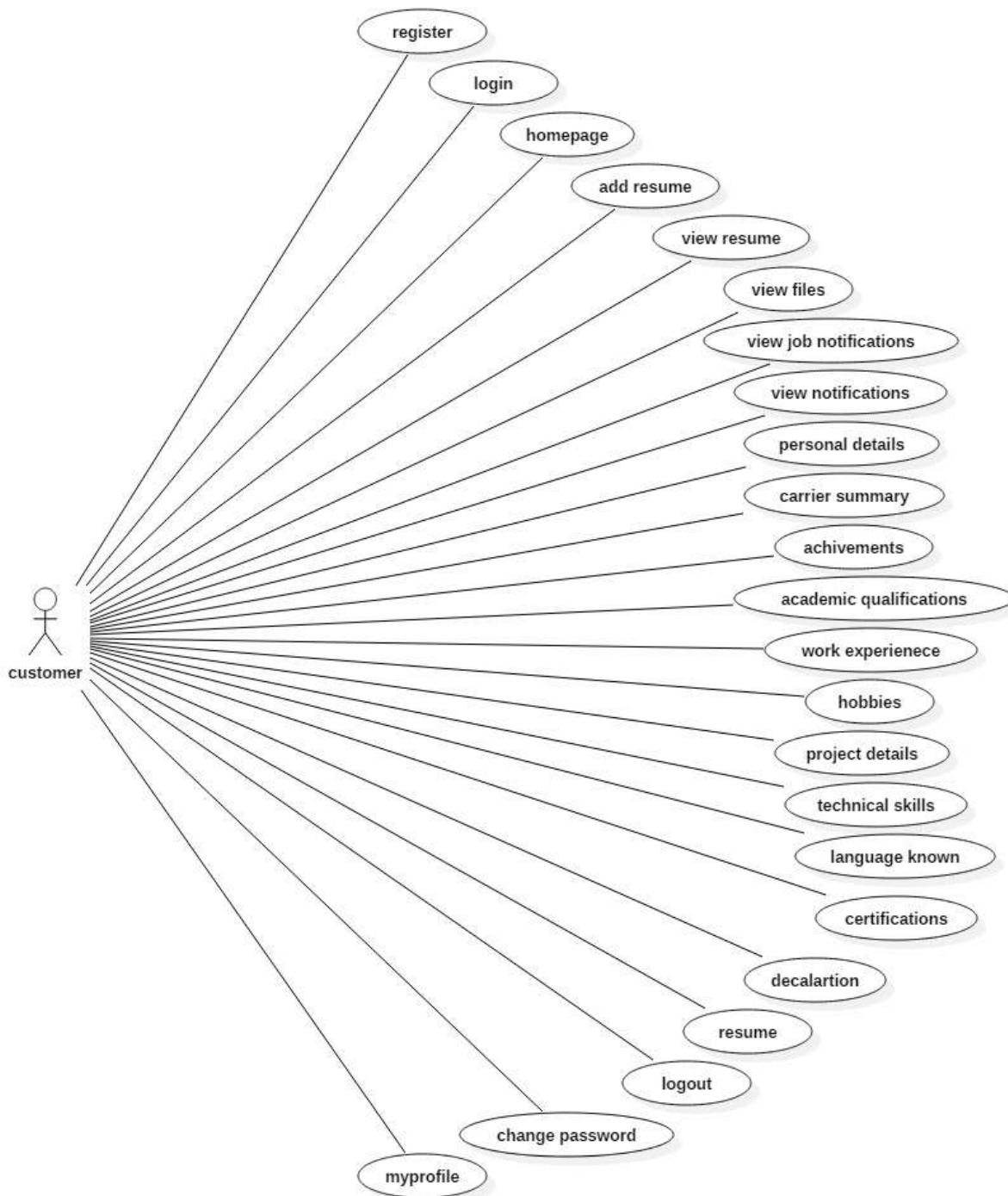
An utilization case outline in the UML is a sort of conduct graph characterized by and made from a Use-case examination. Its motivation is to introduce a graphical diagram of the usefulness given by a framework regarding on-screen characters, their objectives (spoke to as use cases), and any

conditions between those utilization cases. The fundamental motivation behind an utilization case outline is to indicate what framework capacities are performed for which on-screen character. Jobs of the on-screen characters in the framework can be delineated”.

### Admin-Use Case Diagram:



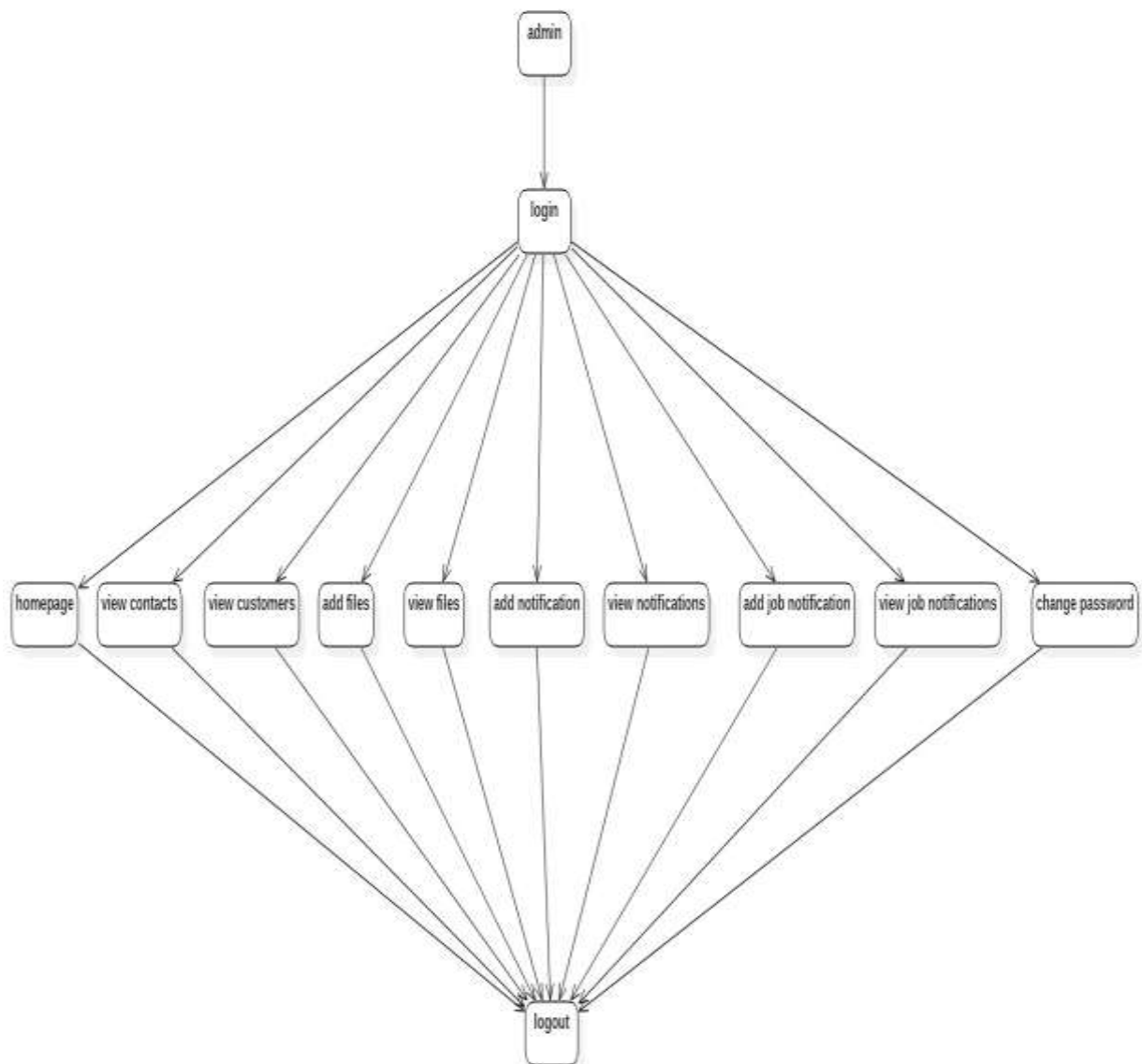
## Customer Use Case Diagram:



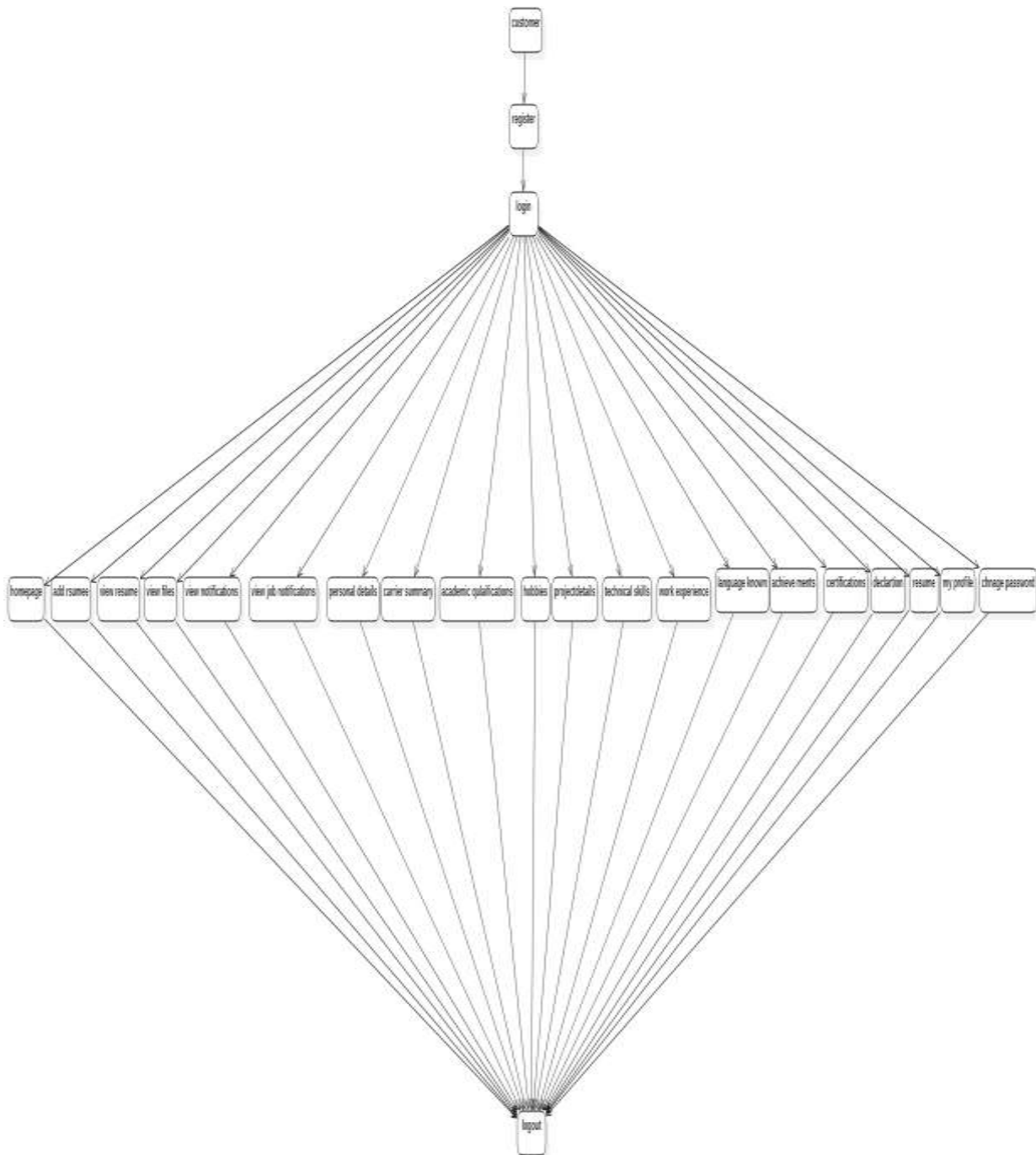
## Activity Diagram

“Movement charts are graphical portrayals of work processes of stepwise exercises and activities with help for decision, emphasis and simultaneousness. In the Unified Modeling Language, movement graphs can be utilized to depict the business and operational bit by bit work processes of parts in a framework. A movement chart demonstrates the general progression of control”.

## Admin-Activity Diagram:

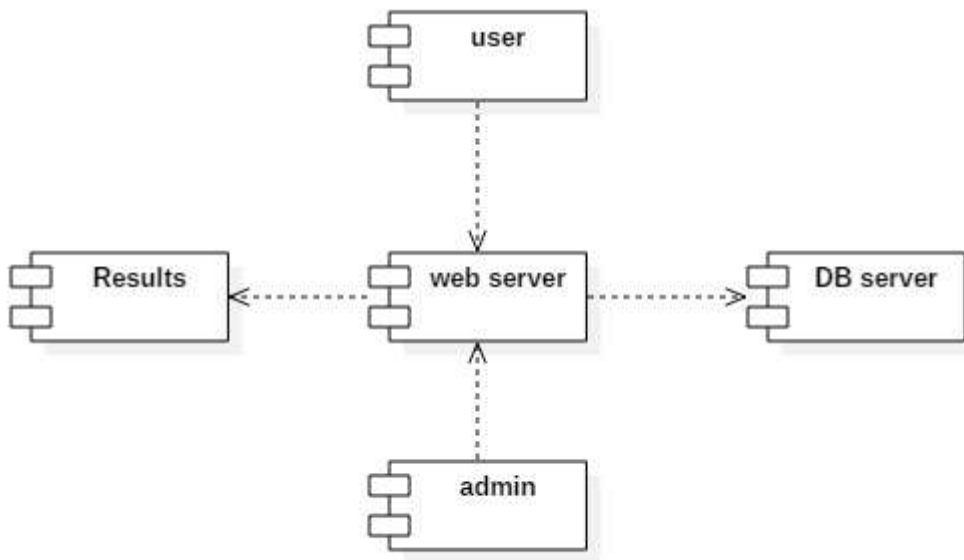


## Customer-Activity Diagram:



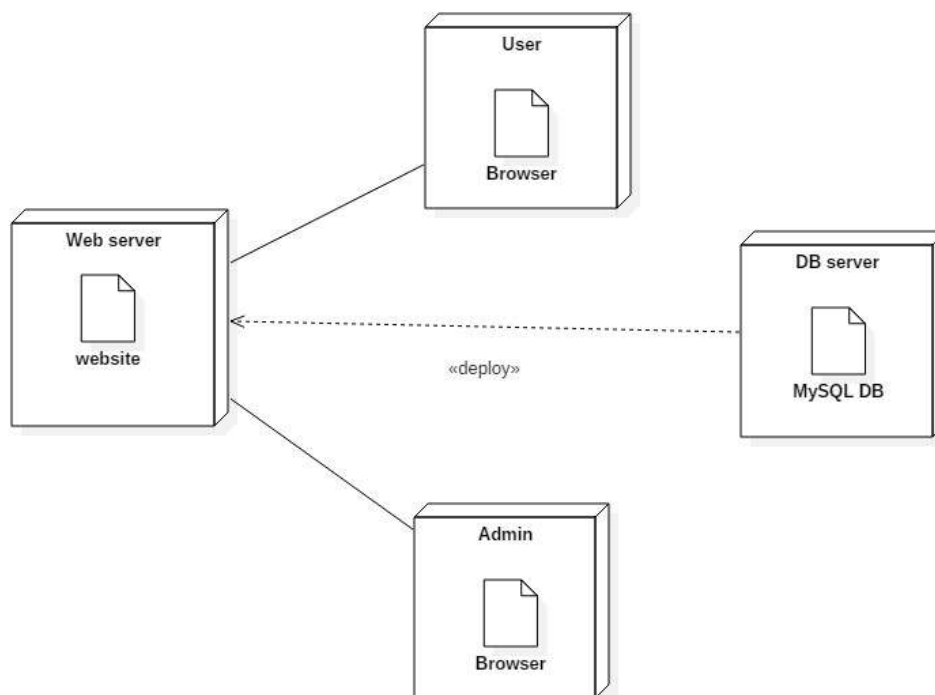
## Component Diagram

The component diagram is a special purpose diagram, which is used to visualize the static implementation view of a system. It represents the physical components of a system, or we can say it portrays the organization of the components inside a system. The components, such as libraries, files, executables, etc.



## Deployment Diagram

A deployment diagram is a UML diagram type that shows the execution architecture of a system, including nodes such as hardware or software execution environments, and the middleware connecting them. Deployment diagrams are typically used to visualize the physical hardware and software of a system

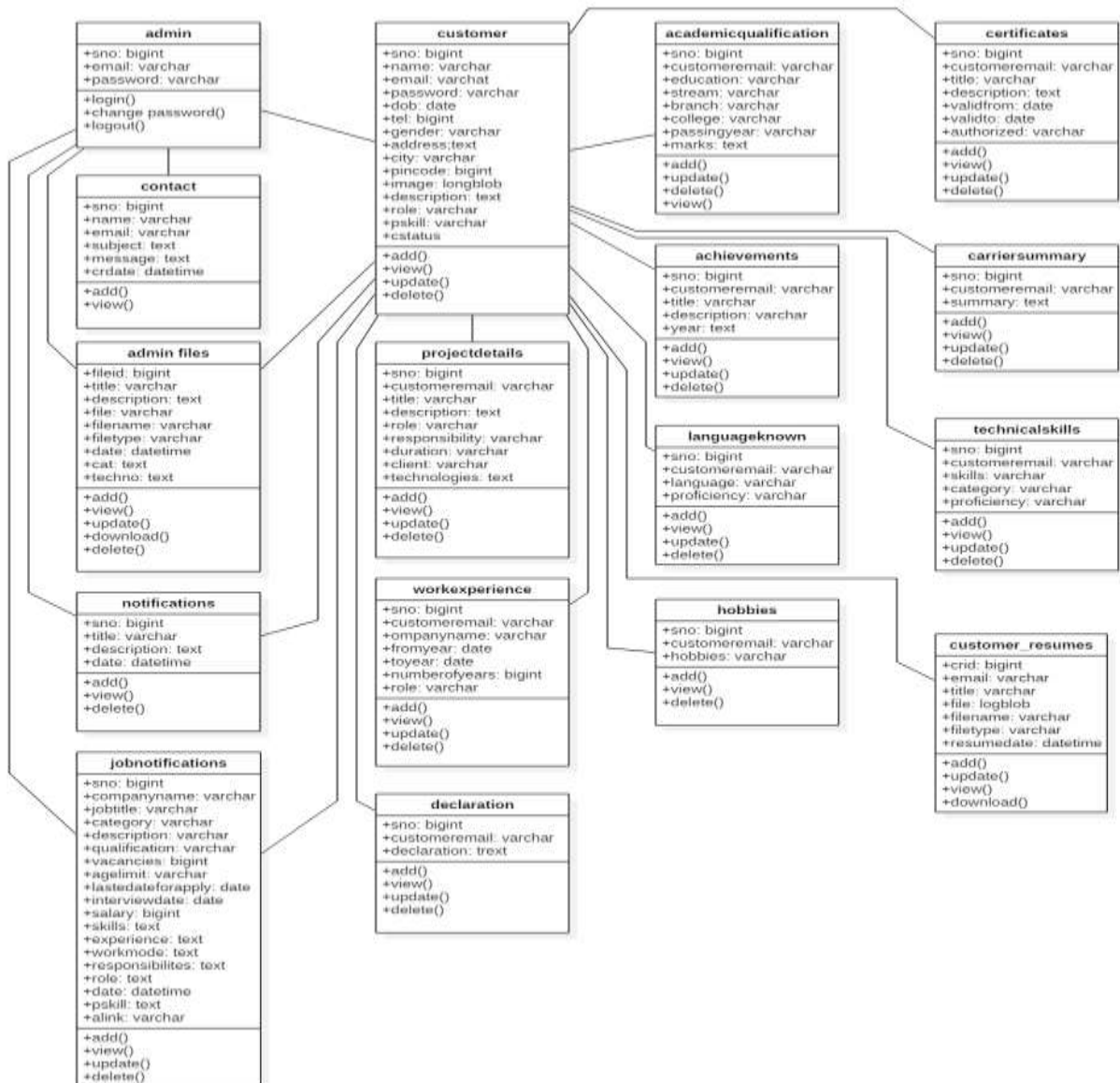


## Class Diagram:

Class diagram is a static diagram. It represents the static view of an application. Class diagram is not only used for visualizing, describing, and documenting different aspects of a system but also for constructing

executable code of the software application.

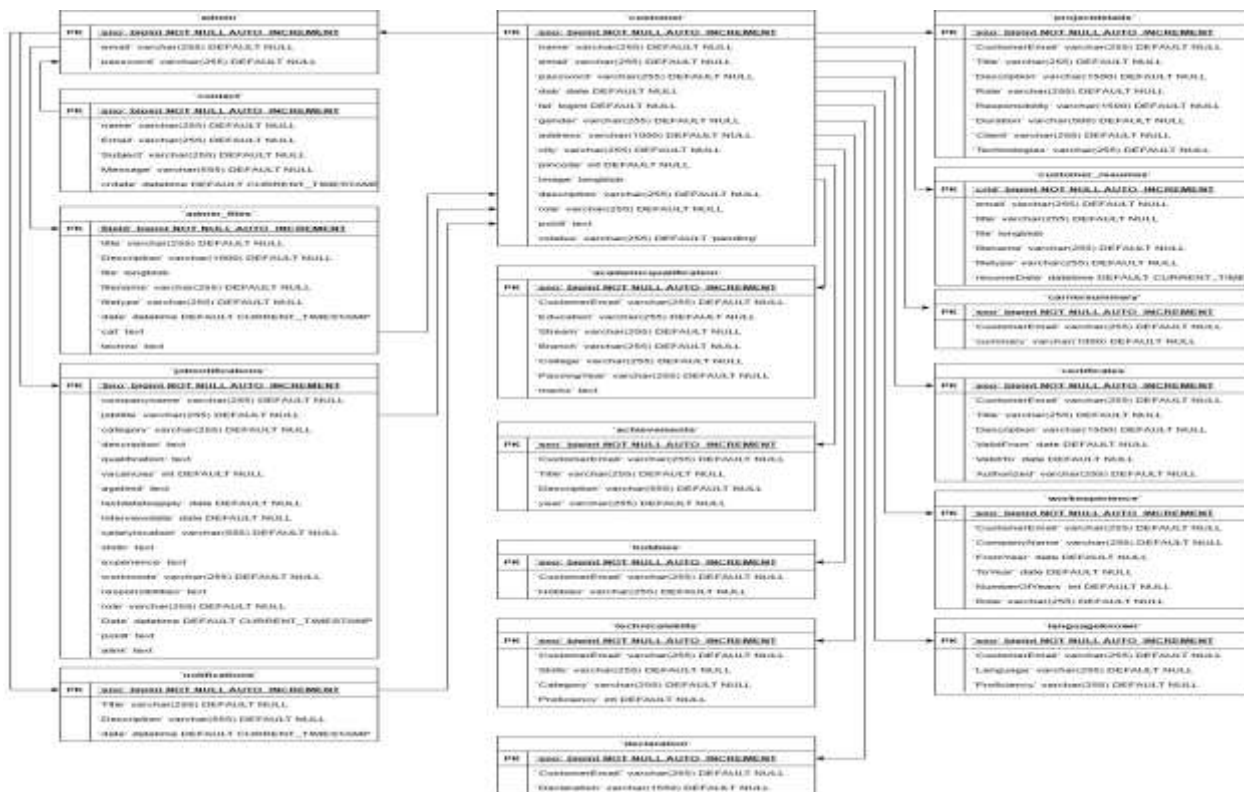
Class diagram describes the attributes and operations of a class and also the constraints imposed on the system. The class diagrams are widely used in the modeling of object oriented systems because they are the only UML diagrams, which can be mapped directly with object-oriented languages.



## ER Diagram

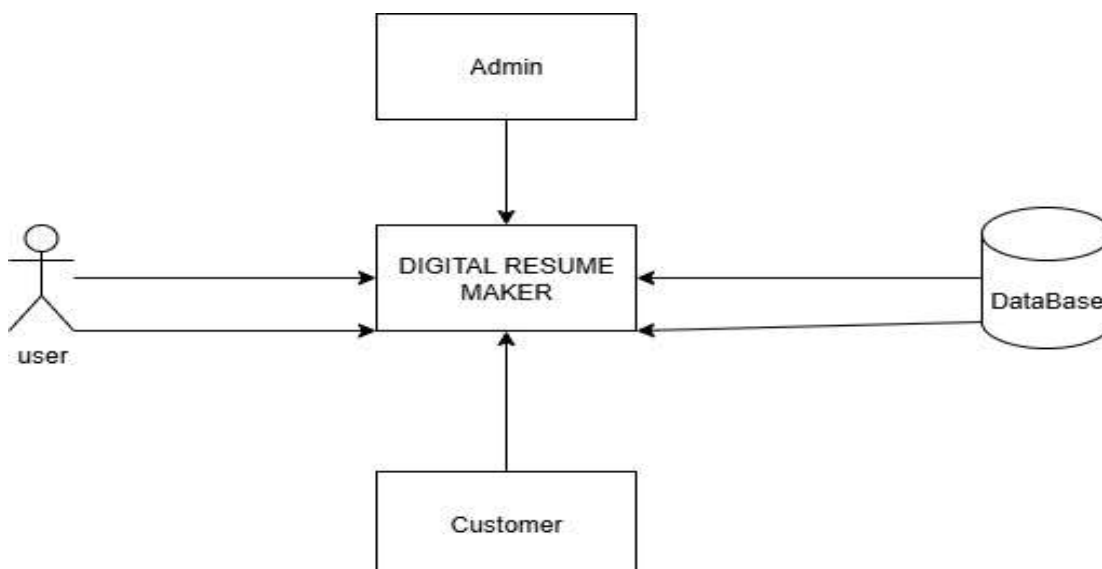
An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research. Also known as ERDs or ER Models, they use a defined set of symbols such as rectangles, diamonds, ovals and connecting lines to depict the interconnectedness

of entities, relationships and their attributes. They mirror grammatical structure, with entities as nouns and relationships as verbs.



## 4.2 Data Flow Diagram

A data-flow diagram is a way of representing a flow of data through a process or a system. The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagram has no control flow — there are no decision rules and no loops. A **data flow diagram** (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows,



# **IMPLEMENTATION**

## **5.1.About Technology : The Python Programming**

### **Language Python**

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. An interpreted language, Python has a design philosophy that emphasizes code readability (notably using whitespace indentation to delimit code blocks rather than curly brackets or keywords), and a syntax that allows programmers to express concepts in fewer lines of code than might be used in languages such as C++ or Java. It provides constructs that enable clear programming on both small and large scales. Python interpreters are available for many operating systems. CPython, the reference implementation of Python, is open source software and has a community-based development model, as do nearly all of its variant implementations. CPython is managed by the non-profit Python Software Foundation. Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

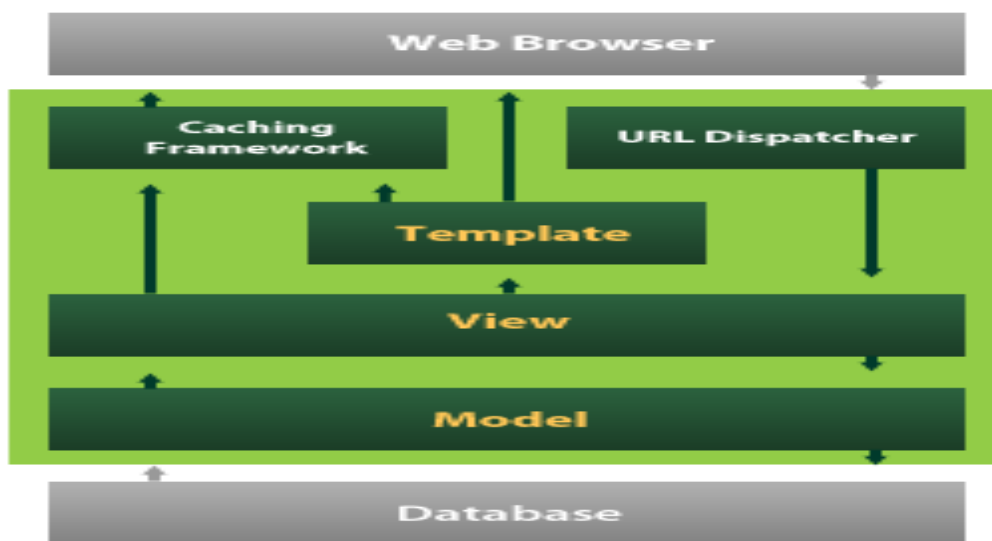
### **Features Of Python:**

- It is general purpose language.
- It is dynamically typed language.
- It is interpreter-based language.
- It is high level language.
- It is portable and scalable.
- It is platform independent language.
- It is object-oriented language.
- It is open-source language and having rich library.
- It supports multi-threading.
- It supports Exception Handling.
- It supports storage facilities like files and database.
- It supports Regular Expressions.
- It is dynamic and distributed language.
- It supports GUI programming
- It is developer friendly language

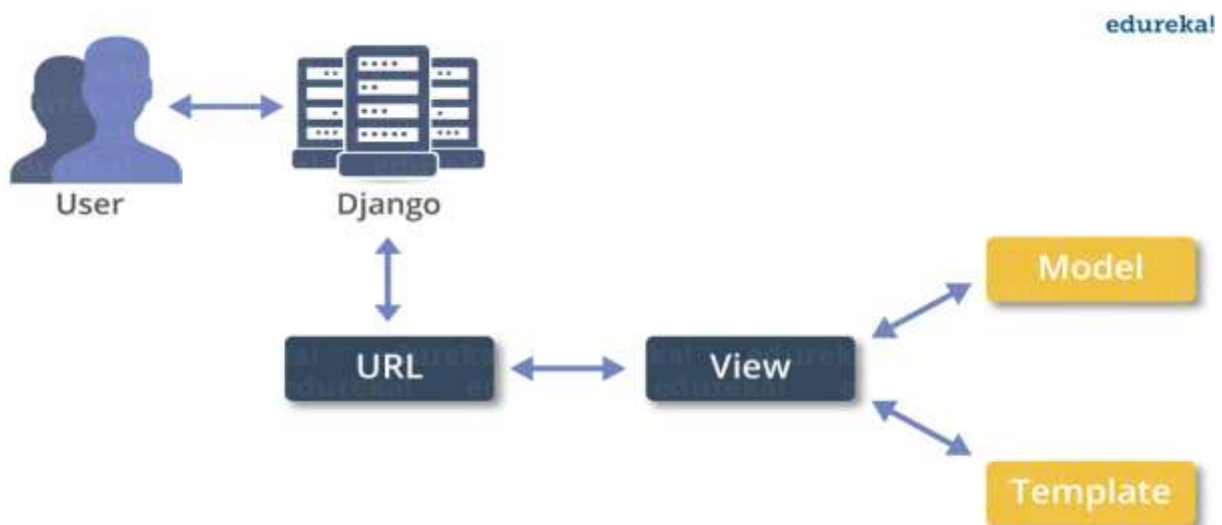
## DJANGO:

Django is a high-level Python Web framework that encourages rapid development and clean, pragmatic design. Built by experienced developers, it takes care of much of the hassle of Web development, so you can focus on writing your app without needing to reinvent the wheel. It's free and open source.

Django's primary goal is to ease the creation of complex, database-driven websites. Django emphasizes reusability and "pluggability" of components, rapid development, and the principle of don't repeat yourself. Python is used throughout, even for settings files and data models.



Django also provides an optional administrative create, read, update and delete interface that is generated dynamically through introspection and configured via admin models



## **HTML:**

Html is used to design the static web pages on web development. Static web page is that kind of web page, in which all the information and material is presented before users as it is exactly stored. Dynamic web page is web page where user can control the operations on website. Html is markup language to develop the web pages. It is basic web technology in web designing and which is used to develop the applications.

Html provides two types of content like input elements which are used to read the input from the user and semantic elements are used to display the data. We have different types of data elements to display the data and working on website. Html have different versions to manage the data efficiently.

## **CSS:**

CSS is a cascading style sheet which will help you to style the webpage . we can apply the effects, styles, font-sizes , animations and others styles to web page. By using css3 we can develop the responsive web pages and we can develop the responsive website. Responsive website can open in the both mobile and browsers effectively.

It is mainly used to apply the color and effects more visibly to the webpage. It is attribute based language and which contains many attribute and value pairs to apply as styles. Css has three versions to apply the styles. It also have different framework like w3.css bootstrap to simplify the css style application.

CSS handles the seem and experience section of a internet page. Using CSS, you can manage the coloration of the text, the fashion of fonts, the spacing between paragraphs, how columns are sized and laid out, what heritage snap shots or colours are used, format designs, versions in show for extraordinary units and display sizes as properly as a range of different effects. CSS is convenient to analyze and recognize however it presents effective manipulate over the presentation of an HTML document. Define fashion sheet regulations immediately along-with the HTML elements the use of fashion attribute. You can practice fashion sheet guidelines without delay to any HTML factor the usage of styleattribute of the applicable tag. This need to be executed solely when you are involved to make a specific trade in any HTML component only. Rules described inline with the factor overrides the guidelines described in an exterior CSS file as nicely as the regulations described in <style> element Define fashion sheet guidelines in header part of the HTML record the usage of <style> tag. If you prefer to practice Style Sheet policies to a single file solely then you can encompass these regulations in header part of the HTML record the usage of <style> tag. Rules described in inside fashion sheet overrides the policies described in an exterior CSS

file. With an exterior fashion sheet, you can trade the seem to be of an whole internet site by using altering simply one file! Each web page need to consist of a reference to the exterior fashion sheet file inner the `<link>` element. The `<link>` aspect goes inner the `<head>` section:

### **Java script:**

It is client-side scripting language which is used to apply the client-side operations and validations to the website. Java script code will be applied in browser itself. It will make the static web page as dynamic webpage in application.

It mainly used for validating the input data, performing minor mathematical operations, dynamic operations, dynamic styles and ajax request to server. Java script is acting as interface between html and php in the time of ajax request. It provides many events and functions to perform the operations on webpage so that static webpage will be converted as dynamic webpage. This code will run browser only and no need of server to install.

Javascript is a dynamic pc programming language. It is light-weight and most often used as a section of internet pages, whose implementations permit client-side script to engage with the consumer and make dynamic pages.

It is first carried out with the aid of Netscape (with assist from Sun Microsystems). JavaScript used to be created with the aid of Brendan Eich at Netscape in 1995 for the reason of permitting code in webpages.

Using HTML we can solely diagram a net web page however you can now not run any good judgment on internet browser like addition of two numbers, take a look at any condition, looping statements (for, while), choice making assertion (if-else) at consumer side. All these are no longer feasible the use of HTML So for function all these assignment at customer aspect you want to use JavaScript.

Comment is nothing however it is a declaration which is no longer show on browser window. it is beneficial to apprehend the which code is written for what purpose. Comments are beneficial in each programming language to supply message. It is used to add statistics about the code, warnings or guidelines so that the cease person or different developer can without problems interpret the code.

Client-side validation, Dynamic drop-down menus, Displaying information and time, Validate person enter in an HTML structure earlier than sending the records to a server, Build varieties that reply to consumer enter besides gaining access to a server, Change the look of HTML files and dynamically

write HTML into separate Windows., Open and shut new windows or frames, Manipulate HTML "layers" which includes hiding, moving, and permitting the person to drag them round a browser window, Build small however entire patron facet packages , Displaying popup home windows and dialog packing containers (like alert dialog box, verify dialog container and instant dialog box), Displaying clocks etc.

## **MYSQL:**

It is a relational database where we are storing the data in the form of relational tables and sql is used to communicate with database called mysql. Its mainly used to store the data and retrieve the data from the user. It is any kind of environment where we are just interested in storing and organizing the data. A Database is a set of data, organized for easy access. Database is the actual data. DBMS is a software and collection of tools that manages the database. Structure data can be defined as any time to press any key on a keyboard it generate from symbol, that signal can be converted into a symbol on the standard output device, such kind of composition is called as structure data. Anything that is basically represented in the form of audio, video and pictures. So total information in the form of binary format. It is non acceptable data from standard input device. It needs a secondary device, it could be anything like scanner, and web camera, mike etc. such kind of data is called as unstructured data. Control files must be protected.

The database device in which the relationships amongst special tables are maintained is referred to as Relational Database Management System. Both RDBMS and DBMS are used to save statistics in bodily database. A relational records mannequin consists of indexes, keys, overseas keys, tables and their relationships with different tables.

## **5.2Sample Source Code**

Templates:

Customer view resume:

```
{% load static %}
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<link rel="stylesheet"
```

```
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
```

```

<style>
body {
  background-color: #f8f9fa;
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  margin: 0;
  padding: 0;
}

.profile-img-passport {
  width: 150px;
  height: 180px;
  object-fit: cover;
  border-radius: 5px;
  border: 1px solid #ccc;
}

.resume-section {
  margin-bottom: 25px;
}

.resume-section h2 {
  border-bottom: 2px solid #dee2e6;
  padding-bottom: 8px;
  margin-bottom: 20px;
  color: #343a40;
}

.progress {
  height: 20px;
  margin-bottom: 10px;
}

.no-print {
  text-align: center;
  margin-top: 20px;
}

```

```
}
```

```
/* A4 sizing for screen */
```

```
#resume-section {  
  background: #fff;  
  width: 794px;  
  min-height: 1123px;  
  margin: 30px auto;  
  padding: 40px;  
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.2);  
  box-sizing: border-box;  
}
```

```
@media print {  
  body {  
    -webkit-print-color-adjust: exact !important;  
    print-color-adjust: exact !important;  
    margin: 0;  
    padding: 0;  
  }  
}
```

```
#resume-section {  
  width: 210mm;  
  min-height: 297mm;  
  margin: 0 auto;  
  padding: 20mm !important;  
  box-shadow: none;  
}
```

```
.no-print, .navbar, .footer, .ftco-section.img {  
  display: none !important;  
}
```

```
body * {  
  visibility: hidden;
```

```

    }

    #resume-section, #resume-section * {
        visibility: visible;
    }

    html, body {
        width: 210mm;
        height: 297mm;
    }
}

</style>
</head>
<body>

<!-- Optional Header -->
{% include 'customer_header.html' %}


<section class="ftco-section ftco-hireme img " style="background-image: url(images/bg_12.jpg)">
    <div class="container">
        <div class="row justify-content-center">
            <div class="col-md-7 ftco-animate text-center">
                <h2>Resume</h2>
            </div>
        </div>
    </div>
</section>


<section class="ftco-section bg-light " >
<div class="container">
<!-- Resume Container -->
<section class="container" id="resume-section">

<!-- Customer Details -->

```

```
{% for customer in customers %}
<div class="d-flex align-items-start mb-4">
  <div>
    
  </div>
  <div>
    <h2 style="color:black">{{ customer.name }}</h2>
    <p><strong>Role:</strong> {{ customer.role }}</p>
    <p><strong>Email:</strong> {{ customer.email }} | <strong>Phone:</strong> {{
customer.number }}</p>
    <p><strong>City:</strong> {{ customer.city }}</p>
    <p><strong>Description:</strong> {{ customer.desc }}</p>
  </div>
</div>
{% endfor %}
```

```
<!-- About Me -->
<div class="resume-section">
  <h2>About Me</h2>
  {% for customer in customers %}
  <ul class="list-unstyled">
    <li><strong>Gender:</strong> {{ customer.gender }}</li>
    <li><strong>Email:</strong> {{ customer.email }}</li>
    <li><strong>Phone:</strong> {{ customer.number }}</li>
  </ul>
  {% endfor %}
</div>
```

```
<!-- Profile Summary -->
<div class="resume-section">
  <h2>Profile Summary</h2>
  <ul>
    {% for summary in summarys %}
    <li>{{ summary.summary }}</li>
    {% endfor %}
  </ul>
```

```

</ul>
</div>

<!-- Education -->
<div class="resume-section">
  <h2>Education</h2>
  {% for education in educations %}
  <p><strong>{{ education.year }} - {{ education.education }}</strong><br>
    {{ education.stream }} | {{ education.college }} | Area: {{ education.branch }}<br>
    Marks/Percentage: {{ education.marks }}</p>
  {% endfor %}
</div>

<!-- Projects -->
<div class="resume-section">
  <h2>Projects</h2>
  {% for project in projects %}
  <p><strong>{{ project.title }}</strong><br>
    Role: {{ project.role }}<br>
    Technologies: {{ project.technologies }}<br>
    Responsibility: {{ project.res }}<br>
    Client: {{ project.client }}<br>
    Description: {{ project.description }}</p>
  {% endfor %}
</div>

<!-- Experience -->
<div class="resume-section">
  <h2>Experience</h2>
  {% for experience in experiences %}
  <p><strong>{{ experience.companyname }}</strong><br>
    Role: {{ experience.role }}<br>
    From: {{ experience.fromyear }} To: {{ experience.toyear }}<br>
    Total Experience: {{ experience.noyears }}</p>
  {% endfor %}

```

</div>

<!-- Skills -->

<div class="resume-section">

<h2>Skills</h2>

{% for skill in skills %}

<p><strong>{{ skill.skill }}</strong> ({{ skill.catagery }})</p>

<div class="progress">

<div class="progress-bar bg-info" role="progressbar" style="width: {{ skill.percentage }}%;">

{{ skill.percentage }}%

</div>

</div>

{% endfor %}

</div>

<!-- Languages -->

<div class="resume-section">

<h2>Languages</h2>

{% for language in languages %}

<p><strong>{{ language.language }}</strong></p>

<div class="progress">

<div class="progress-bar bg-success" role="progressbar" style="width: {{ language.percent }}%;">

{{ language.percent }}%

</div>

</div>

{% endfor %}

</div>

<!-- Hobbies -->

<div class="resume-section">

<h2>Hobbies</h2>

<ul>

{% for hobbie in hobbies %}

<li>{{ hobbie.hobbies }}</li>

```

        {% endfor %}
    </ul>
</div>

<!-- Achievements -->
<div class="resume-section">
    <h2>Achievements</h2>
    <ul>
        {% for a in achievements %}
            <li><strong>{{ a.title }}</strong> | {{ a.description }} | {{ a.ayear }}</li>
        {% endfor %}
    </ul>
</div>

<!-- Certifications -->
<div class="resume-section">
    <h2>Certifications</h2>
    <ul>
        {% for c in certifications %}
            <li><strong>{{ c.title }}</strong> |
                {{ c.description }} |
                Valid: {{ c.validfrom }} to {{ c.validto }} |
                Authorized by: {{ c.authorized }}</li>
        {% endfor %}
    </ul>
</div>

<!-- Declaration -->
<div class="resume-section">
    <h2>Declaration</h2>
    <ul>
        {% for dec in decs %}
            <li>{{ dec.dec }}</li>
        {% endfor %}
    </ul>

```

```

</div>
</section>

<!-- Buttons (Not Printed) -->
<div class="no-print">
  <button class="btn btn-primary" onclick="window.print()"> 🖨 Print / Download Resume</button>
</div>
</div></section>
<!-- Footer -->
{% include 'footer.html' %}
</body>
</html>

```

#### Admin view users:

```

<!DOCTYPE html>
<html lang="en">
<head>
  {% load static %}
  <meta charset="UTF-8">
  <title>View Users</title>
  <link rel="stylesheet" href="{% static 'css/style.css' %}">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">
  {% include 'admin_header.html' %}
</head>
<body>

<section class="ftco-section ftco-hireme img" style="background-image: url('{% static
'images/resume1212.jpg' %}'); height:600px;">
  <div class="container">
    <div class="row justify-content-center">
      <div class="col-md-7 text-center">
        <h2 style="margin-top:120px">View Users</h2>
      </div>

```

```

        </div>
    </div>
</section>

<section class="ftco-section bg-light">
<div class="container">
    <div class="row">
        {% for user in users %}
        <div class="col-lg-4 col-md-6 mb-4">
            <div class="card h-100 shadow-sm text-center p-2">
                {% if user.image %}
                
                {% else %}
                
                {% endif %}

                <div class="card-body">
                    <h4 class="text-primary">
                        <i class="fas fa-user text-dark"></i>&nbsp;&nbsp;&nbsp;{{ user.name }}
                    </h4>
                    <p><i class="fas fa-envelope text-dark"></i>&nbsp;&nbsp;&nbsp;{{ user.email }}</p>
                    <p><i class="fas fa-phone text-dark"></i>&nbsp;&nbsp;&nbsp;{{ user.number }}</p>
                    <p><i class="fas fa-venus-mars text-dark"></i>&nbsp;&nbsp;&nbsp;{{ user.gender }}</p>
                    <p><i class="fas fa-city text-dark"></i>&nbsp;&nbsp;&nbsp;{{ user.city }}</p>
                    <p><i class="fas fa-user-tag text-dark"></i>&nbsp;&nbsp;&nbsp;{{ user.role }}</p>
                    <p><i class="fas fa-quote-left text-
dark"></i>&nbsp;&nbsp;&nbsp;<strong>Objective:</strong> {{ user.desc|truncatewords:20 }}</p>

                    {% if user.status == 'On Hold' %}
                    <a href="{% url 'accept_customer' id=user.id %}" class="btn btn-success btn-sm me-
2">

```

```

        <i class="fas fa-check"></i>
    </a>
    <a href="{% url 'reject_customer' id=user.id %}" class="btn btn-danger btn-sm">
        <i class="fas fa-times"></i>
    </a>
    {% elif user.status == 'Accepted' %}
        <button class="btn btn-warning btn-sm" disabled><i class="fas fa-thumbs-
up"></i>&nbsp;&nbsp;&nbsp;Accepted</button>
    {% else %}
        <button class="btn btn-danger btn-sm" disabled><i class="fas fa-
ban"></i>&nbsp;&nbsp;&nbsp;Rejected</button>
    {% endif %}
</div>
</div>
</div>
{% endfor %}
</div>
</div>
</section>
<br><br>
</body>
</html>
{% include 'footer.html' %}

```

Views:

```
import mimetypes
```

```
from wsgiref.util import FileWrapper
```

```
from django.shortcuts import render,redirect,HttpResponse
```

```
from django.template.loader import get_template
```

```
from xhtml2pdf import pisa
```

```
from Resume_maker_app.models import
```

```
Contact,Customer,Education,Project,Skills,Language,Experience,Career,Hobbies,Dec, Admin,
```

```

Notification, Achievements, Certifications, Upload_Resume, Job_Notification, Add_Files
from Resume_maker_app.forms import
ContactForms, CustomerForms, EducationForms, ProjectForms, SkillForms, LanguageForms, Experienc
eForms, CareerForms, HobbiesForms, DecForms, NotificationForm, AchievementsForms,
CertificationsForms, Upload_ResumesForms, Job_NotificationsForms, Add_FilesForms
from io import BytesIO
import os

```

```

# Create your views here.

```

```

def index(request):
    return render(request, "index.html", {})

```

```

def about(request):
    return render(request, "about.html", {})

```

```

def customer_page(request):
    return render(request, "customer_page.html", {})

```

```

def resume(request):
    email=request.session['email']
    customers=Customer.objects.filter(email=email)
    project=Project.objects.filter(customer_id=customers[0].id)
    education=Education.objects.filter(customer_id=customers[0].id)
    skill=Skills.objects.filter(customer_id=customers[0].id)
    language=Language.objects.filter(customer_id=customers[0].id)
    experience=Experience.objects.filter(customer_id=customers[0].id)
    summary=Career.objects.filter(customer_id=customers[0].id)
    achievements=Achievements.objects.filter(customer_id=customers[0].id)
    certifications=Certifications.objects.filter(customer_id=customers[0].id)
    hobbies=Hobbies.objects.filter(customer_id=customers[0].id)

```

```

    dec=Dec.objects.filter(customer_id=customers[0].id)
    return render(request, "resume.html",
{"customers":customers,"projects":project,"educations":education,"skills":skill,"languages":language
,"experiences":experience,"summarys":summary,"hobbies":hobbies,"decs":dec,"achievements":achie
vements,"certifications":certifications })

```

```

def resume_templates(request):
    return render(request, "resume_templates.html")

```

```

def resume_style1(request):
    email = request.session['email']
    customer = Customer.objects.filter(email=email).first()
    context = {
        "customers": [customer],
        "projects": Project.objects.filter(customer_id=customer.id),
        "educations": Education.objects.filter(customer_id=customer.id),
        "skills": Skills.objects.filter(customer_id=customer.id),
        "languages": Language.objects.filter(customer_id=customer.id),
        "experiences": Experience.objects.filter(customer_id=customer.id),
        "summarys": Career.objects.filter(customer_id=customer.id),
        "achievements": Achievements.objects.filter(customer_id=customer.id),
        "certifications": Certifications.objects.filter(customer_id=customer.id),
        "hobbies": Hobbies.objects.filter(customer_id=customer.id),
        "decs": Dec.objects.filter(customer_id=customer.id),
    }
    return render(request, "resume_style1.html", context)

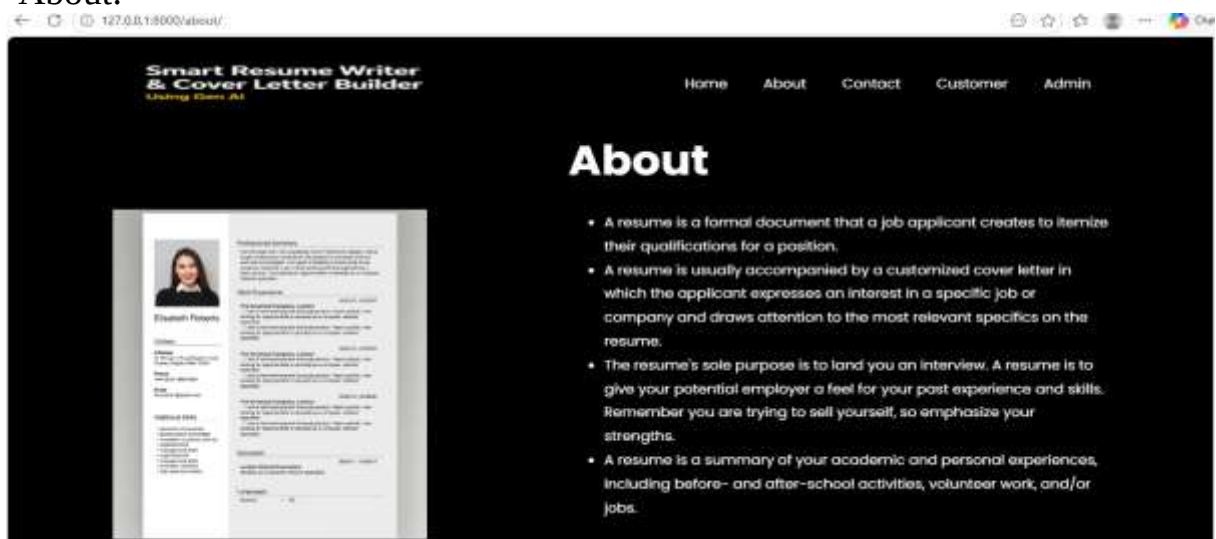
```

## 5.3.SCREEN SHORTS:

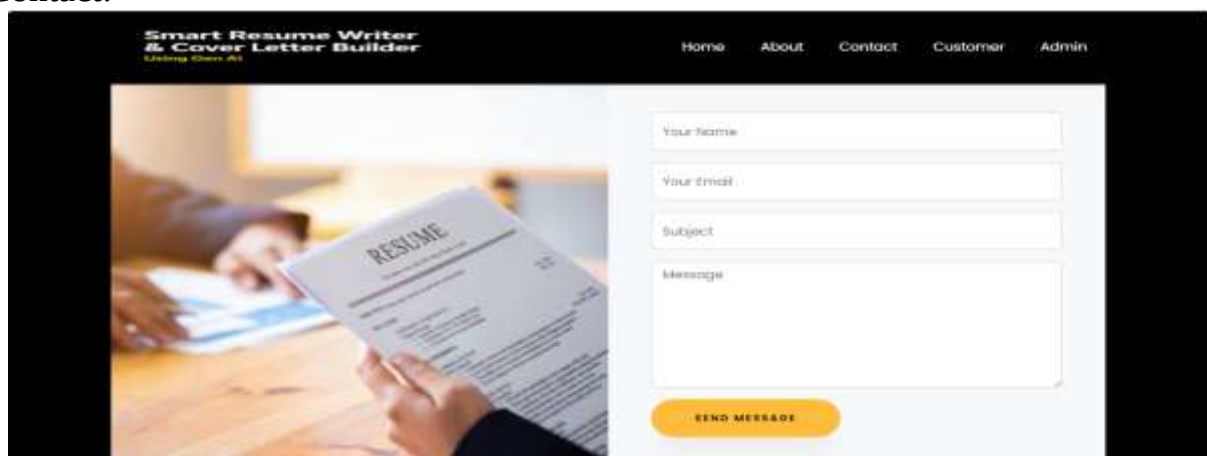
Index:



About:



Contact:



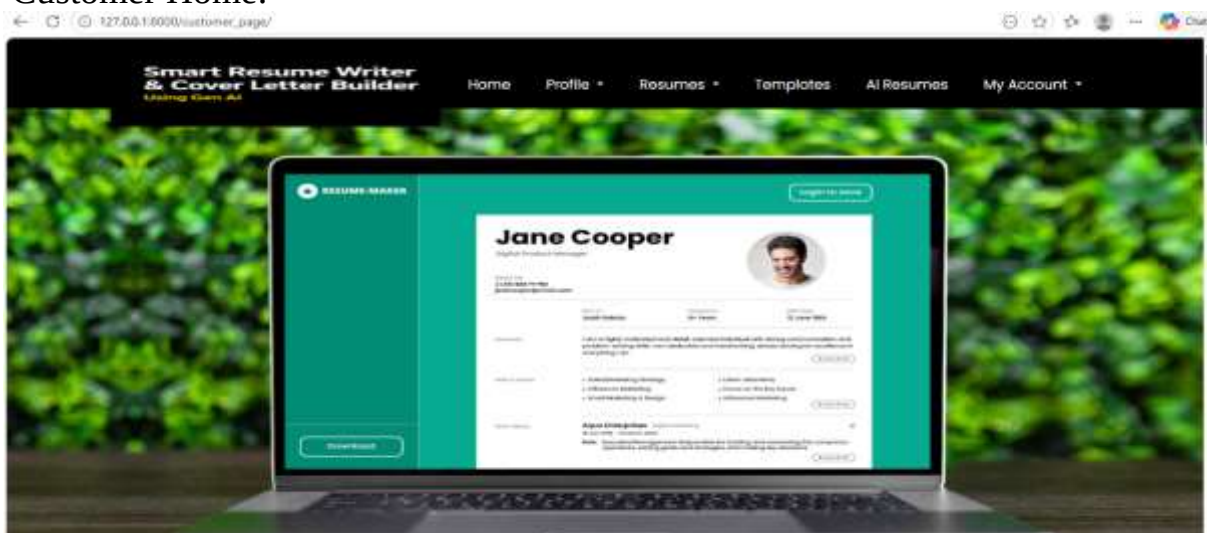
## Customer Registration:

The screenshot shows the 'Smart Resume Writer & Cover Letter Builder' website with the URL '127.0.0.1:8000/customer\_regpage/'. The page features a dark header with navigation links: Home, About, Contact, Customer, and Admin. The main content area is a light blue form with the following fields: Email (labeled 'Your Email'), Password (labeled 'Your Password'), Name (labeled 'Your Name'), Gender (radio buttons for Male, Female, Other), and Role (labeled 'Your role').

## Customer Login:

The screenshot shows the 'Smart Resume Writer & Cover Letter Builder' website with the URL '127.0.0.1:8000/customer/'. The page features a dark header with navigation links: Home, About, Contact, Customer, and Admin. The main content area is a light blue form titled 'Customer Login' with the following fields: Email (labeled 'Your Email'), Password (labeled 'Your Password'), a LOGIN button, a FORGOT PASSWORD button, and a link for 'User not exist? Register'.

## Customer Home:



## Customer Profile Details:

The screenshot shows the 'Customer Profile Details' page. The header includes the application name 'Smart Resume Writer & Cover Letter Builder' and navigation links: Home, Profile, Resumes, Templates, AI Resumes, and My Account. The profile card displays the following information:

- S Bharath**
- Gender: male
- Role: Associate Engineer
- Mobile: 9123456789
- City: Anantapur
- Email: badrinath630379@gmail.com
- Career Objective: Career Objective5

An 'EDIT' button is located at the bottom right of the profile card.

## Carrier Summary:

The screenshot shows the 'Summary Details' page. The header is identical to the previous page. The main content area features a decorative banner with the text 'Summary Details' and an illustration of a desk with a laptop, a cup, and documents. Below the banner, the 'Your Career Objective' section displays the email 'badrinath630379@gmail.com' and a description of experience: '-> Developed and maintained web applications using Python and Django framework Designed and implemented RESTful APIs for seamless frontend-backend integration Created and managed database models using Django ORM implemented user authentication and authn'. 'EDIT' and 'DELETE' buttons are at the bottom of this section.

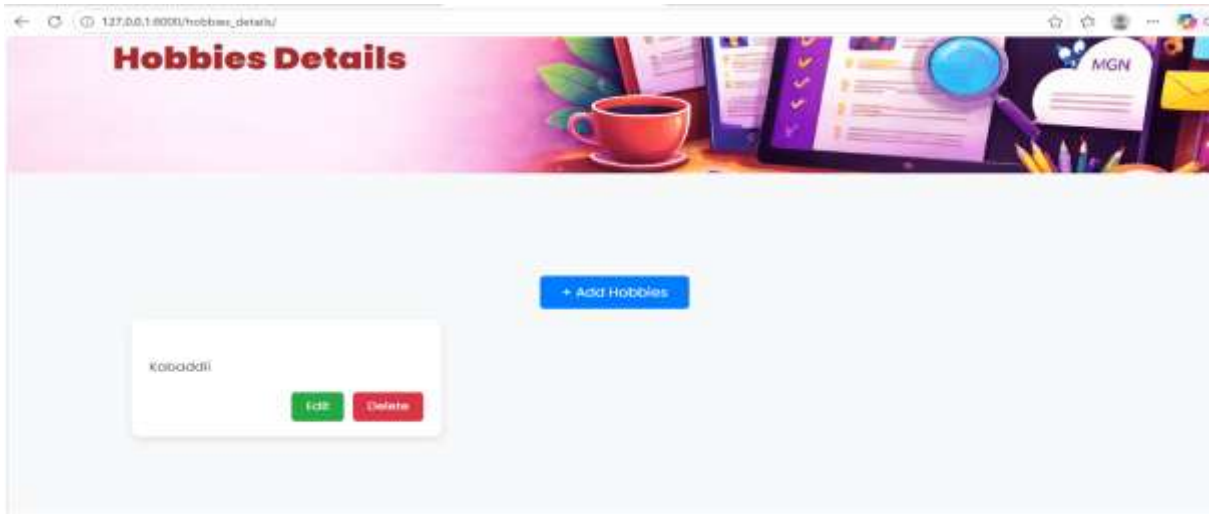
## Educational Details:

The screenshot shows the 'Education Details' page. The header is identical to the previous pages. The main content area displays a table for educational qualifications:

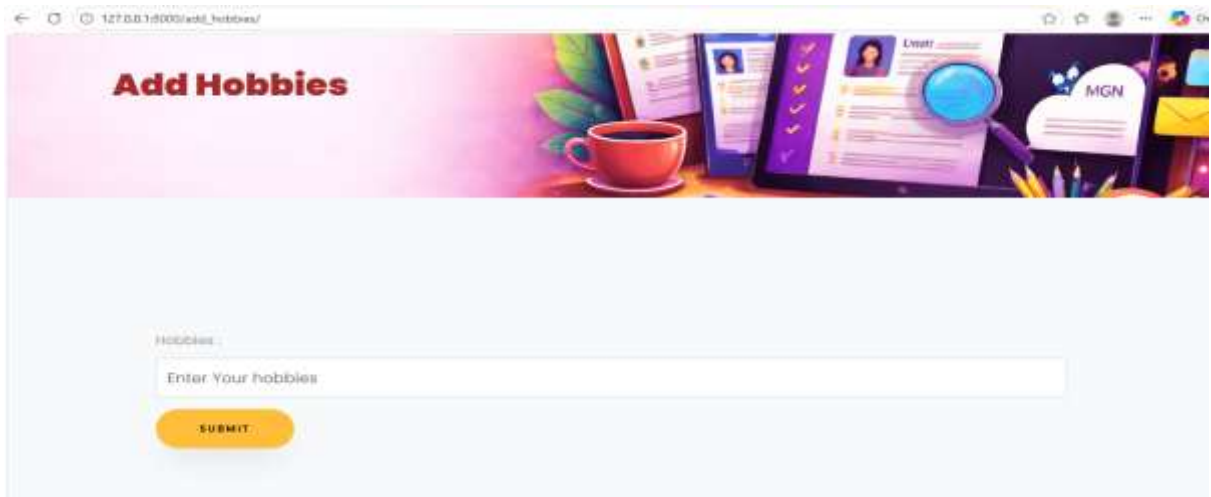
| Bachleor's Degree in BSC  |                     |              | EDIT | DELETE |
|---------------------------|---------------------|--------------|------|--------|
| Badr Degree College       |                     |              |      |        |
| Email                     | Education           | Stream       |      |        |
| badrinath630379@gmail.com | Bachleor's Degree   | BSC          |      |        |
| Branch/Area               | College/School      | Passing Year |      |        |
| Hyderabad                 | Badr Degree College | 2021         |      |        |
| Marks/Percentage          |                     |              |      |        |
| 73%                       |                     |              |      |        |

An 'Add Another Qualification' button is located at the bottom center of the page.

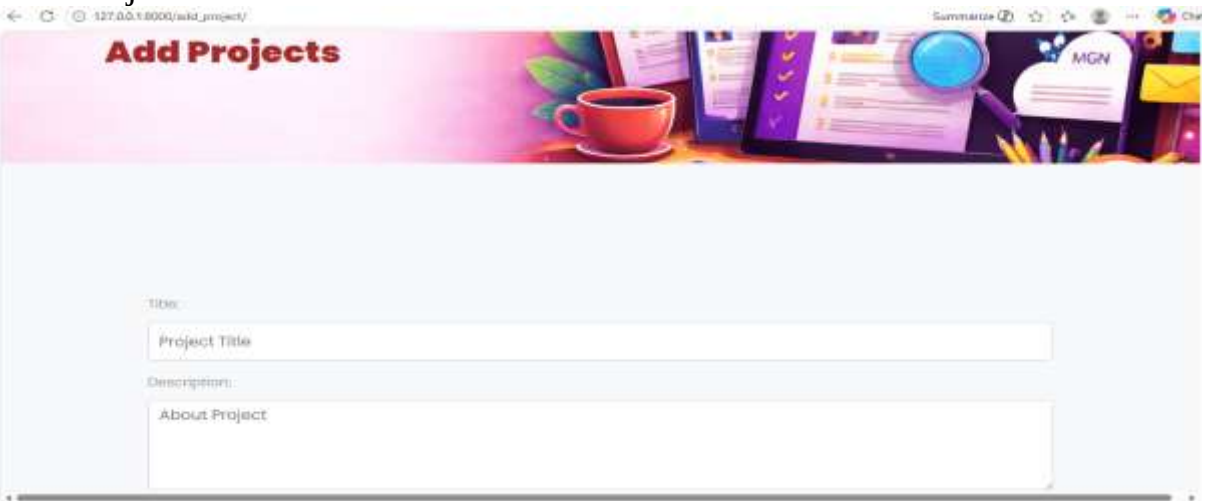
Hobbies view:



Add Hobbies:



Add Project Details:



## Project Details:

Smart Resume Writer & Cover Letter Builder  
Using Open AI

Home Profile Resumes Templates AI Resumes My Account

Add Project

**Skill Shoppy**  
P Sateesh Kumar

**Role**  
Associate Engineer

**Responsibilities**  
This Projects is a Ecomerce web Application project. In this project anybody will register and login and

**Description**  
This projects is web based

**ProjectS Shoppy**  
P Sateesh Kumar

**Role**  
Software Engineer

**Responsibilities**  
This Projects is a Ecomerce web Application project. In this project anybody will register and login and

**Description**  
This Projects is e-commerce web

## Add Skills:

Smart Resume Writer & Cover Letter Builder  
Using Open AI

Home Profile Resumes Templates AI Resumes My Account

Add Skill

**Add Skills**

Skill:  
Your Skills

Category:  
Technical

Save

## Skills Details:

Smart Resume Writer & Cover Letter Builder  
Using Open AI

Home Profile Resumes Templates AI Resumes My Account

+ Add Skill

**Python** (Technical)  
Proficiency: 100%  
Email: bsdnash0037@gmail.com  
Edit Delete

**Selenium** (Technical)  
Proficiency: 75%  
Email: bsdnash0037@gmail.com  
Edit Delete

127.0.0.1:8000/add\_language/

Summarize

# Add Languages



Language:

Percentage/proficient:

SUBMIT

127.0.0.1:8000/add\_experience/

Add Experiences

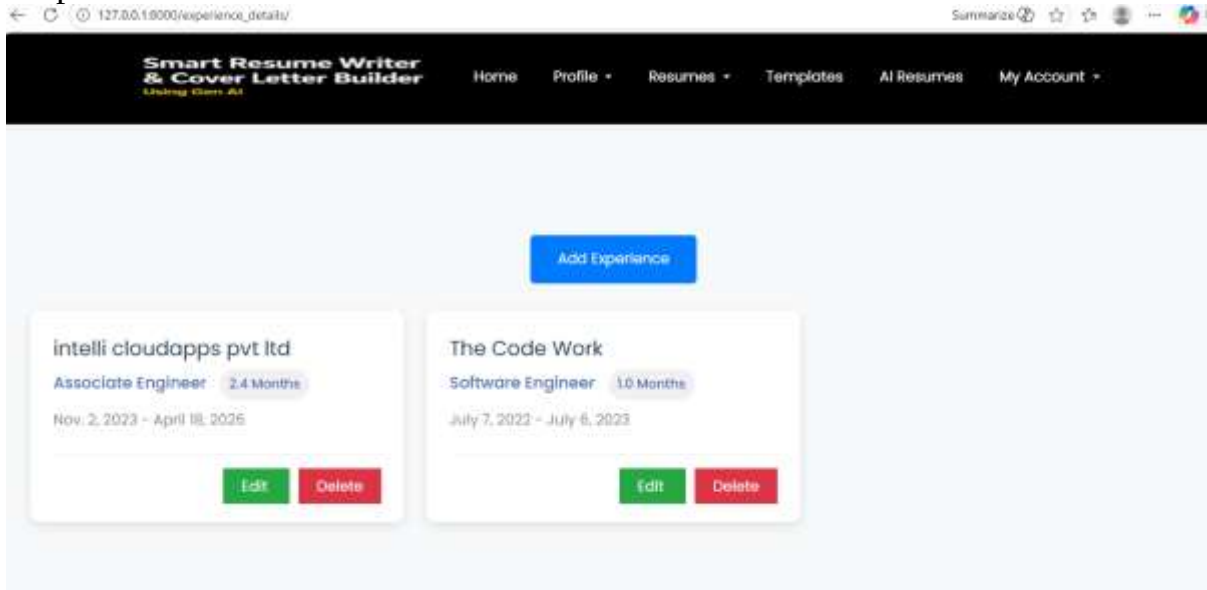


Company Name:

From Date:

To Date:

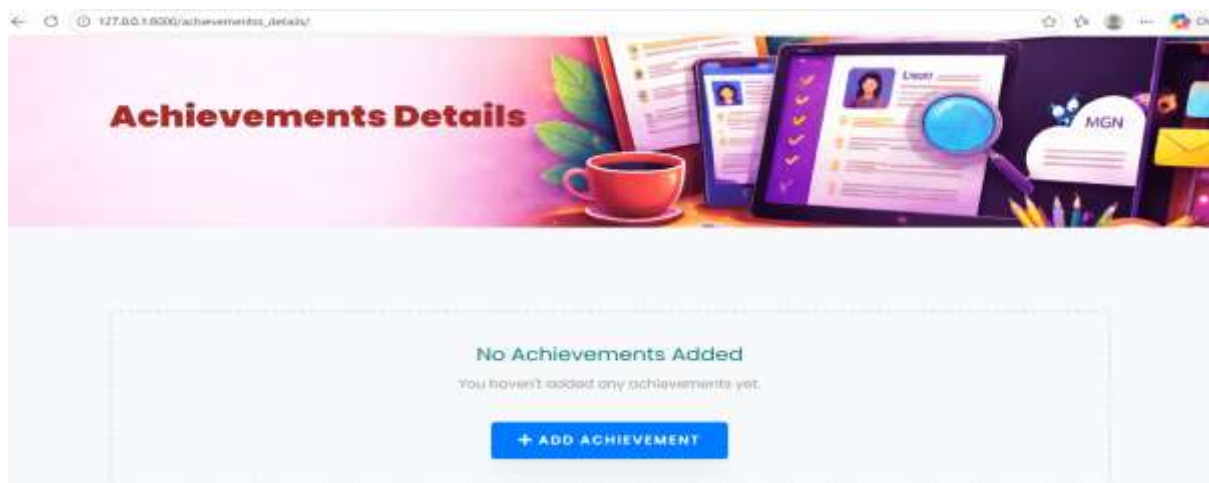
## Experience Details:



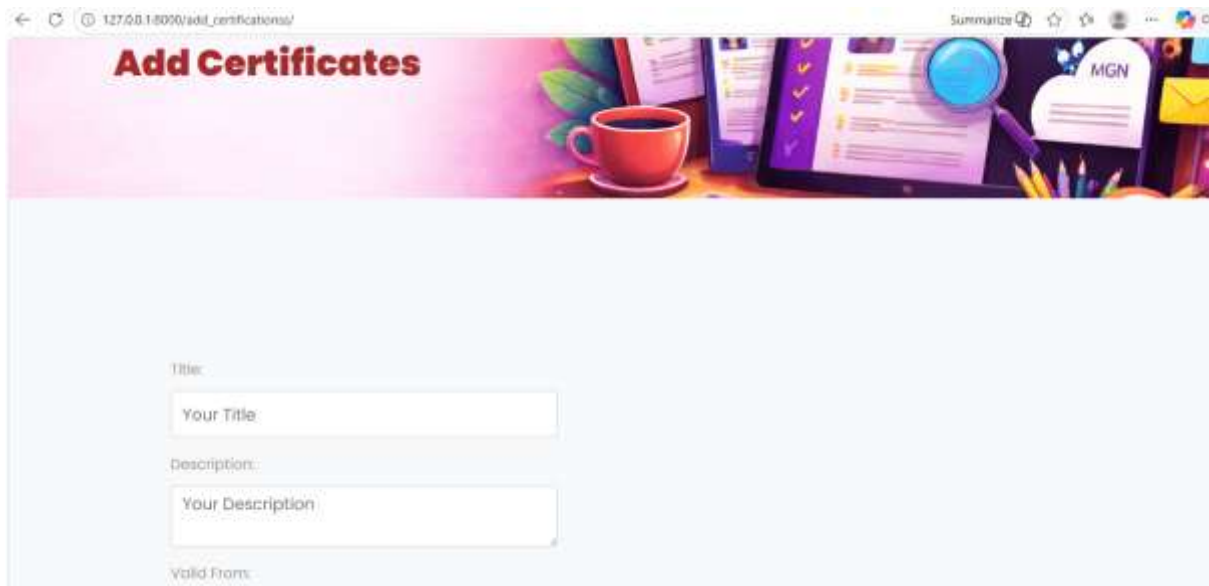
## Add Achievements:



## Achievements Details:



## Add Certificates:



127.0.0.1:8090/add\_certificates/

### Add Certificates

Title:

Description:

Valid From:

## Certificate Details:



127.0.0.1:8090/certificate\_details/

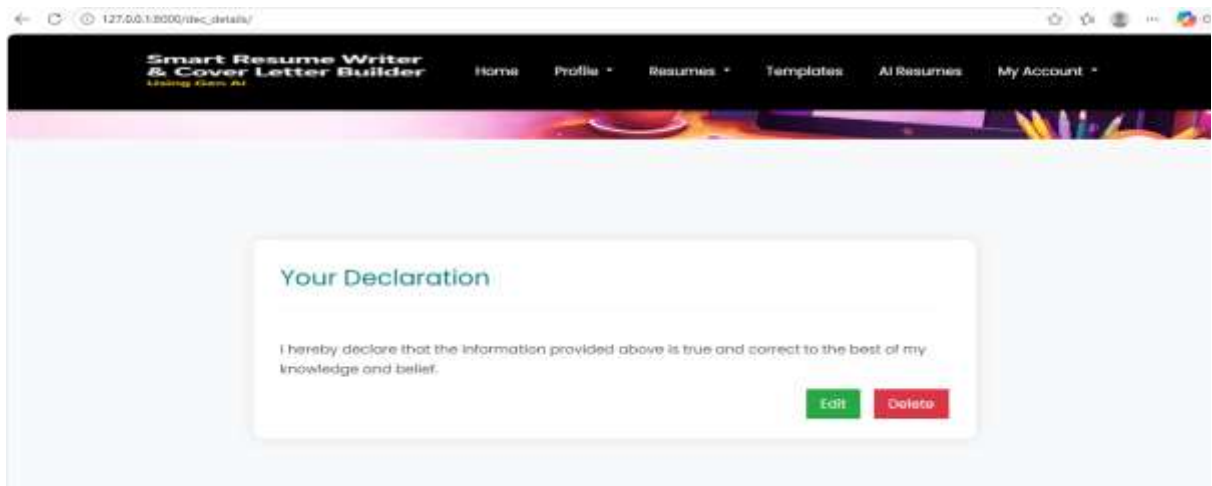
### Certificate Details

No Certificates Found

You haven't added any certificates yet

[+ Add Certificate](#)

## Declaration Details:



127.0.0.1:8090/dec\_details/

### Smart Resume Writer & Cover Letter Builder

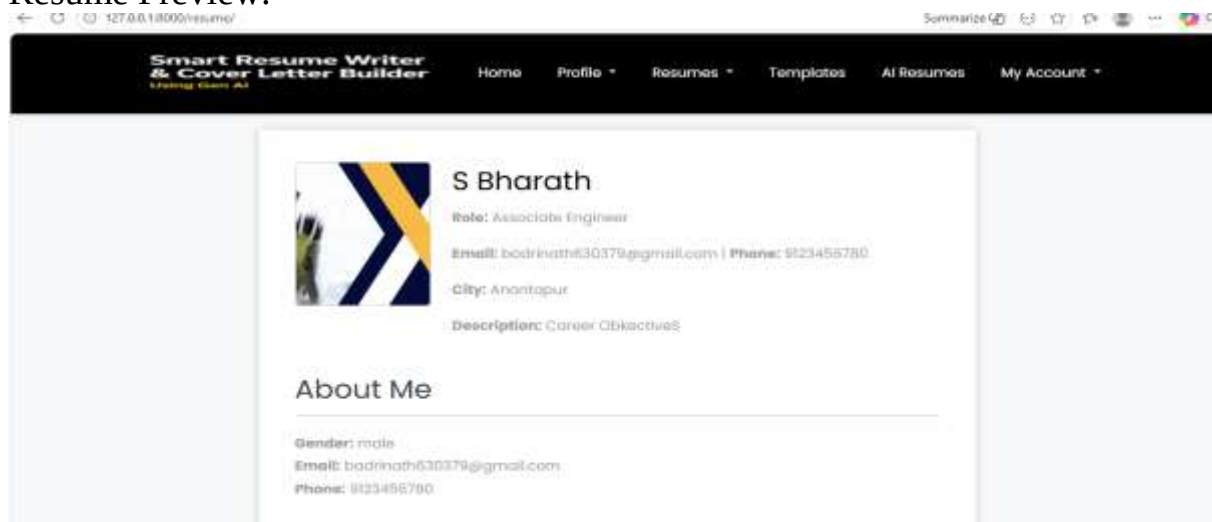
Home Profile Resumes Templates AI Resumes My Account

#### Your Declaration

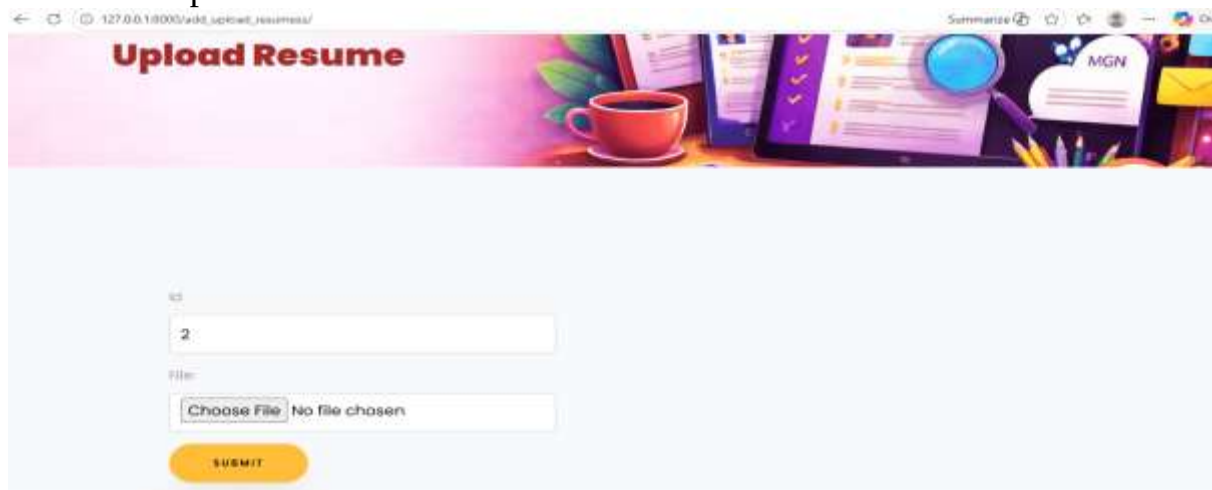
I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

[Edit](#) [Delete](#)

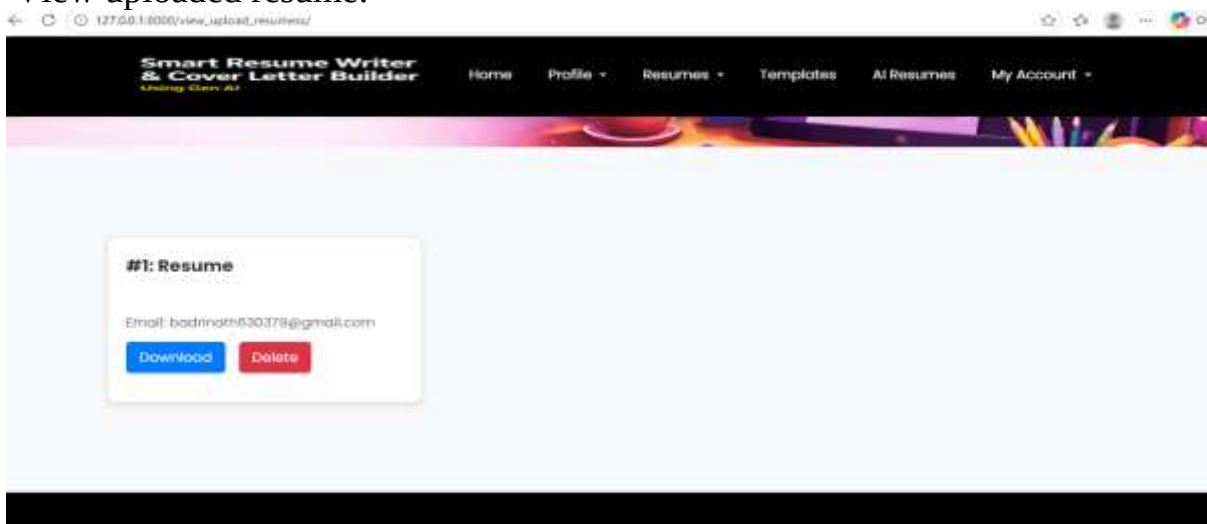
## Resume Preview:



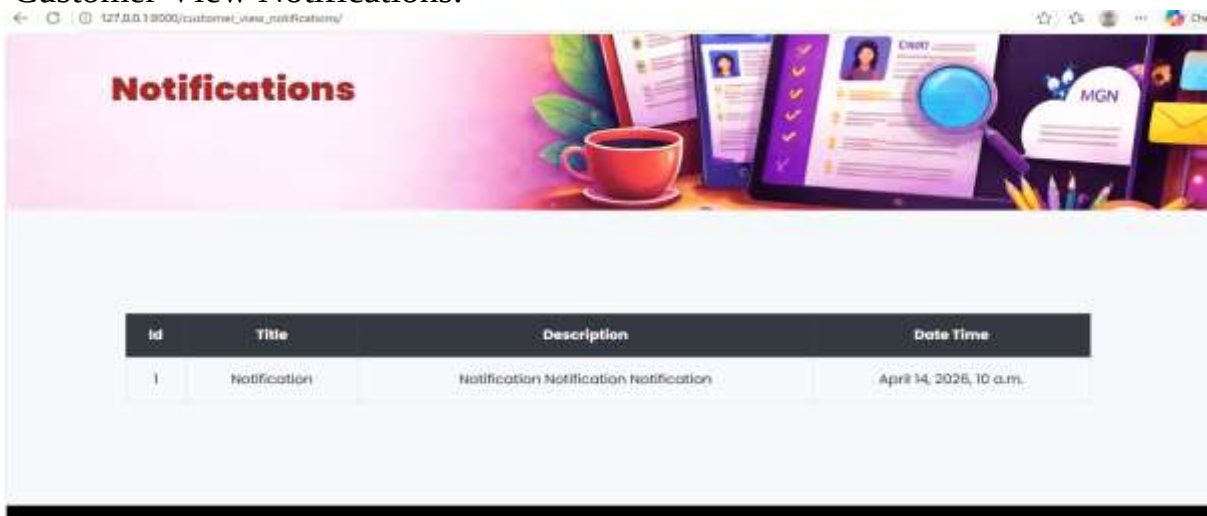
## Customer Upload Resume:



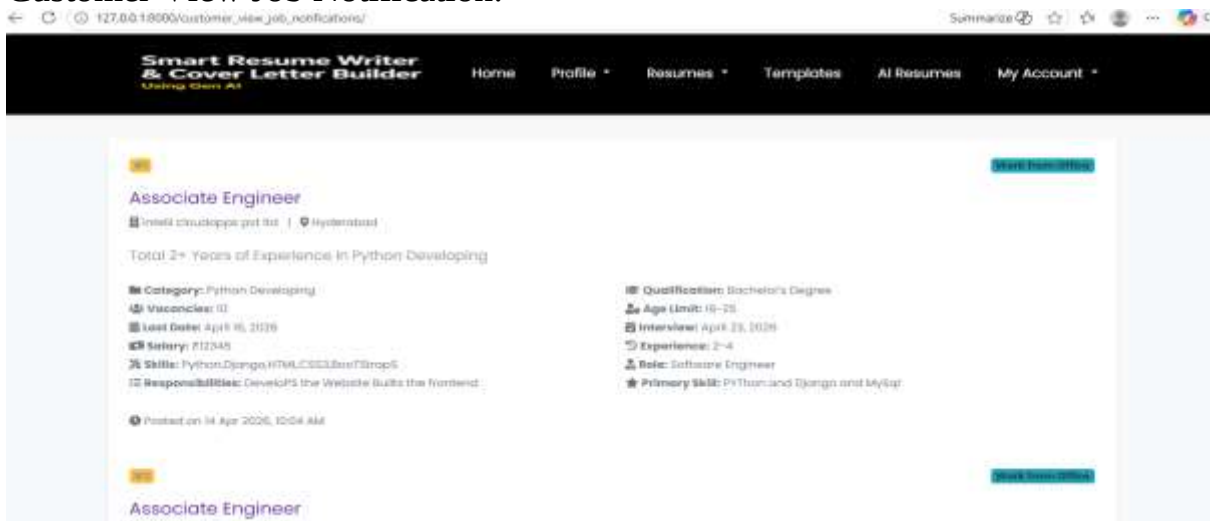
## View uploaded resume:



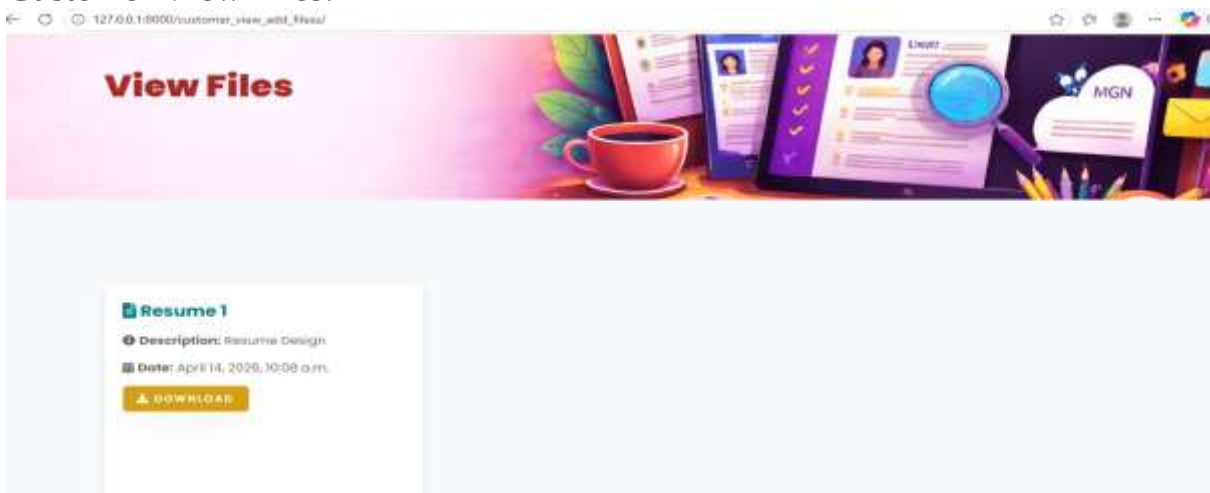
## Customer View Notifications:



## Customer View Job Notification:



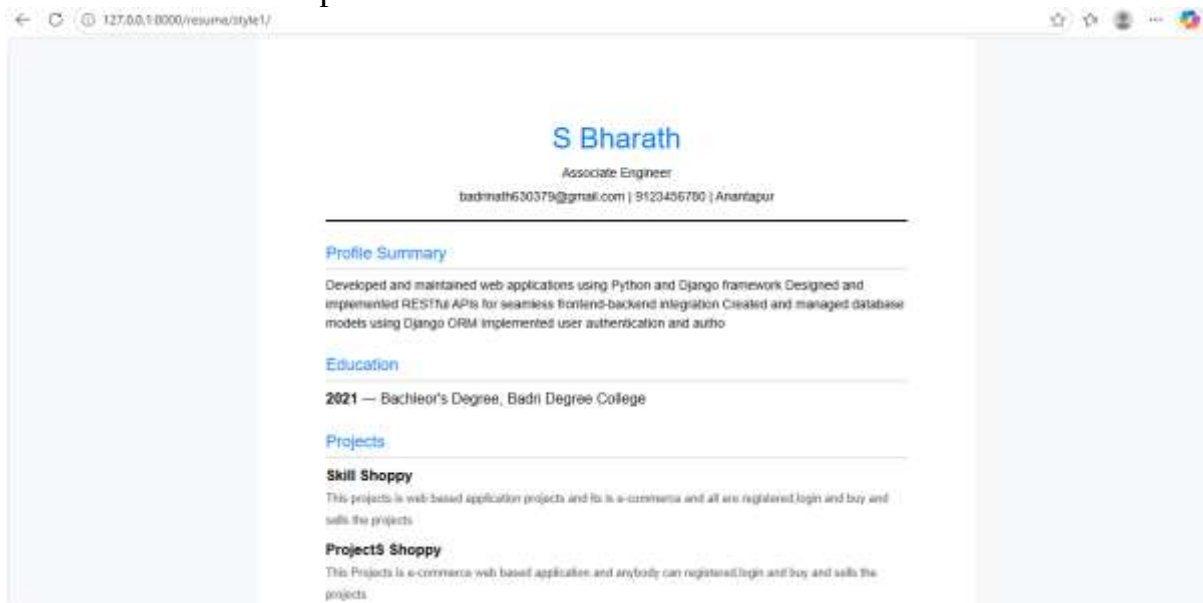
## Customer View Files:



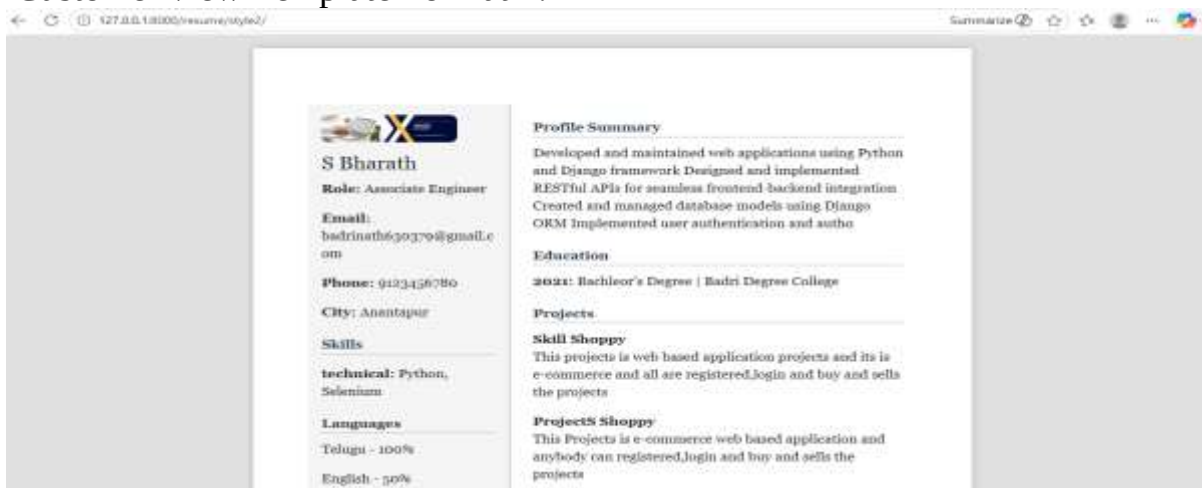
## Customer View Templates Format:



## Customer view Template format-1:



## Customer view Template Format 2:



### Customer View Template Format 3:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8000/resume/style/3'. The page features a profile for 'S Bharath', an Associate Engineer. The profile includes contact information: Email (badrinath630379@gmail.com), Phone (9123456789), and City (Anantapur). A profile summary follows, detailing experience with Python, Django, RESTful APIs, and database models. Below this, the education section lists a 2021 Bachelor's Degree from Sachin Degree College. The projects section includes 'Skill Shoppy' and 'Project5 Shoppy', both described as web-based applications. The page is styled with a light blue background and a white profile card.

**S Bharath**  
Role: Associate Engineer  
Email: badrinath630379@gmail.com  
Phone: 9123456789  
City: Anantapur

**Profile Summary**  
Developed and maintained web applications using Python and Django framework. Designed and implemented RESTful APIs for seamless frontend-backend integration. Created and managed database models using Django ORM. Implemented user authentication and authorization.

**Education**  
2021: Bachelor's Degree, Sachin Degree College

**Projects**  
**Skill Shoppy**  
This project is a web-based application project and its e-commerce and all are registered, login and buy and sell the products.  
**Project5 Shoppy**  
This project is a e-commerce web-based application and includes all the features like login and buy and sell the products.

### Customer Change Password:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8000/customer\_password/'. The page has a black header with the text 'Smart Resume Writer & Cover Letter Builder Using Gen AI' and navigation links: Home, Profile, Resumes, Templates, All Resumes, and My Account. The main content area is a light blue form with fields for Email (badrinath630379@gmail.com), Old Password (Your Old Password), and New Password (Your New Password). A yellow 'SUBMIT' button is at the bottom.

Smart Resume Writer & Cover Letter Builder  
Using Gen AI

Home Profile Resumes Templates All Resumes My Account

Email:  
badrinath630379@gmail.com

Old Password:  
Your Old Password

New Password:  
Your New Password

SUBMIT

### Admin Login:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8000/admin\_login/'. The page features a colorful header banner with the text 'Admin Login' and an illustration of a laptop, a cup of coffee, and various documents. Below the banner is a white login form with fields for Email (Your Email) and Password (Your Password), and a yellow 'LOGIN' button.

Admin Login

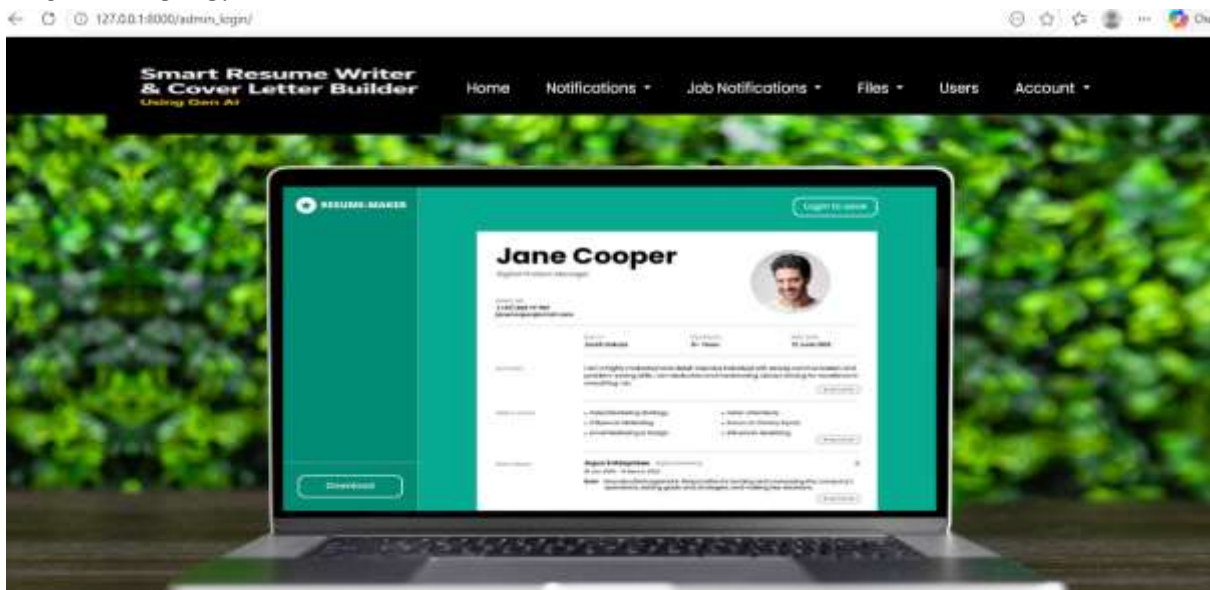
Admin Login

Email:  
Your Email

Password:  
Your Password

LOGIN

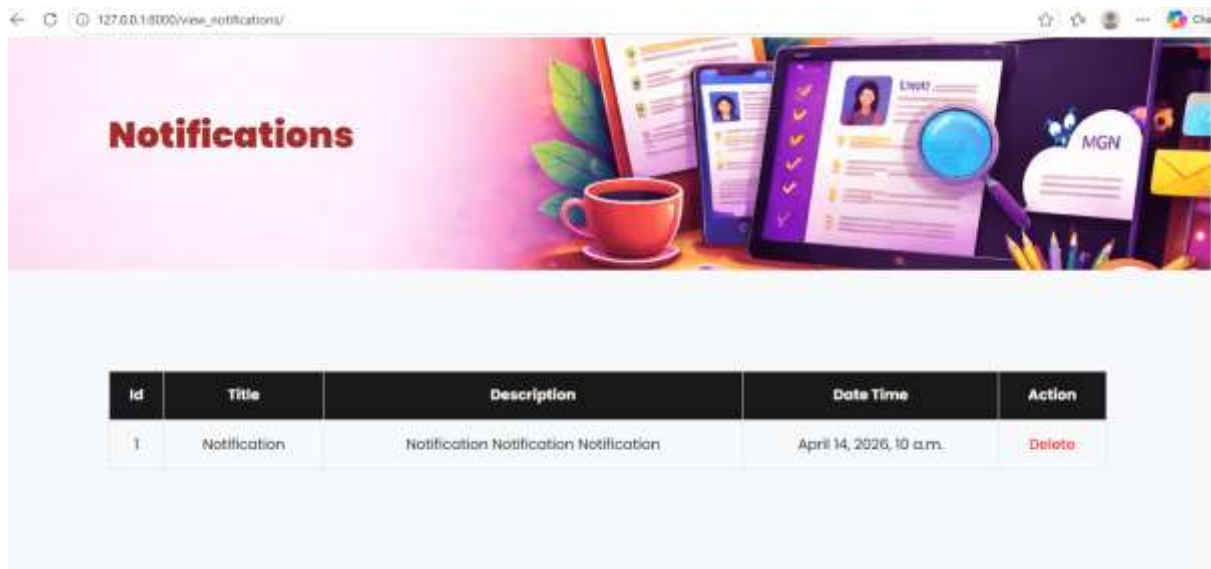
## Admin Home:



## Admin Add Notifications:



## Admin View Notifications:



## Admin Add Job Notifications:



**Add Job Notifications**

Company Name:

Emails:

Qualifications:

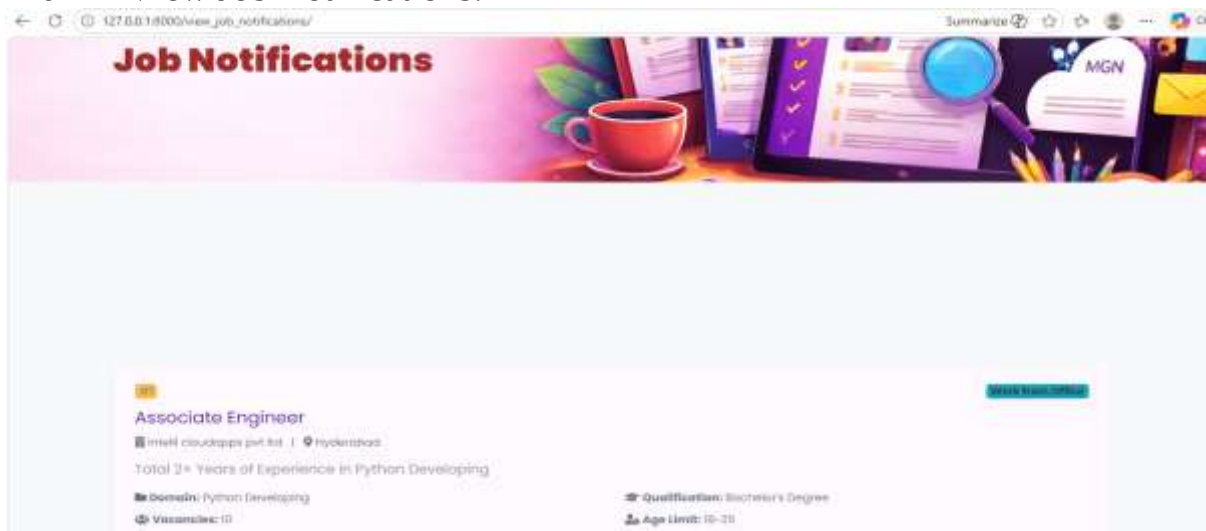
Title:

Description:

Vacancies:

[Submit](#)

## Admin View Job Notifications:



**Job Notifications**

**Associate Engineer**

Intel cloudapps pvt ltd | Hyderabad

Total 2+ Years of Experience in Python Developing

Domain: Python Developing

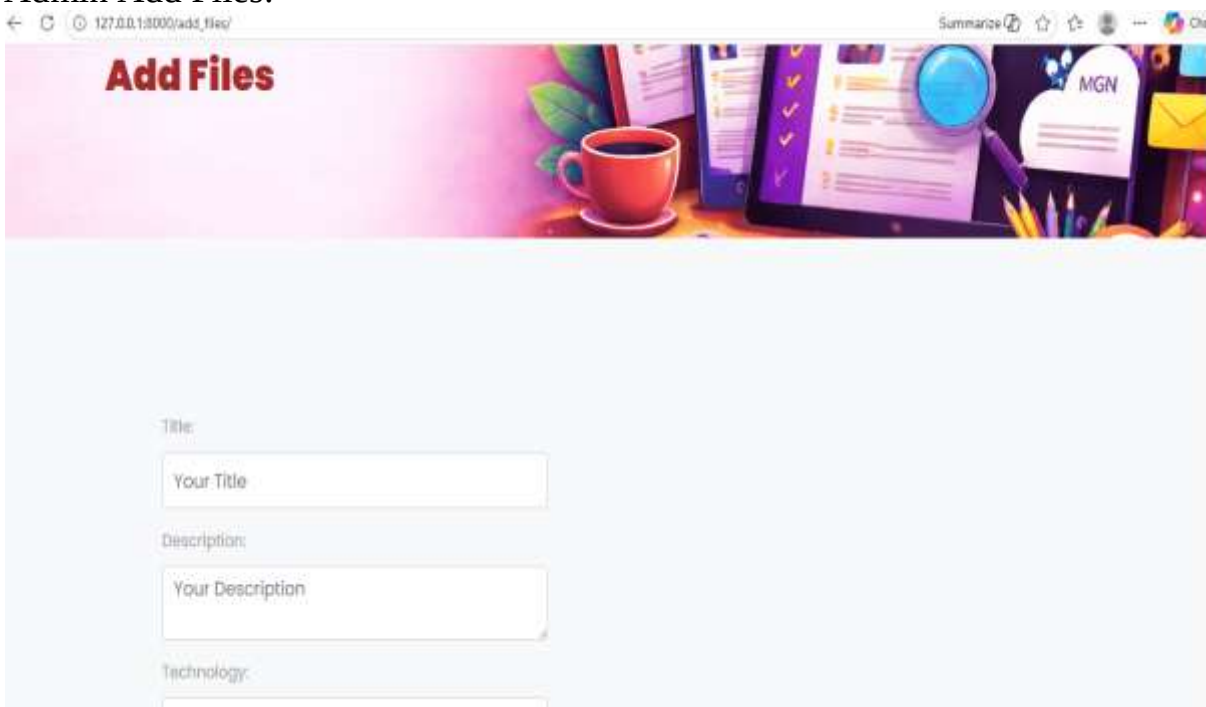
Vacancies: 10

Qualification: Bachelor's Degree

Age Limit: 18-35

[View More Details](#)

## Admin Add Files:



**Add Files**

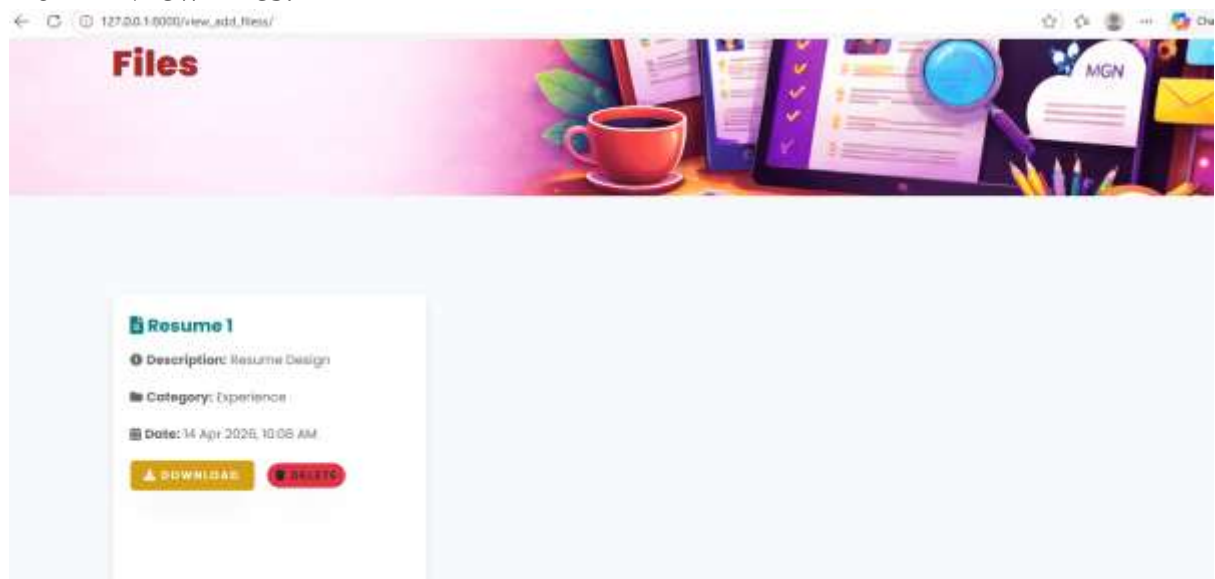
Title:

Description:

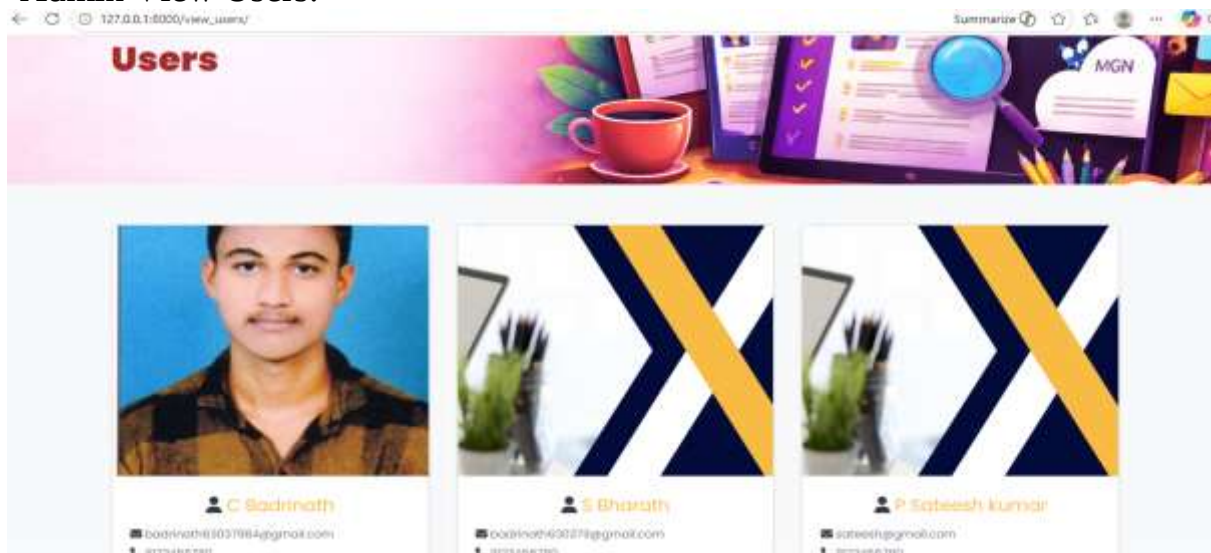
Technology:

[Submit](#)

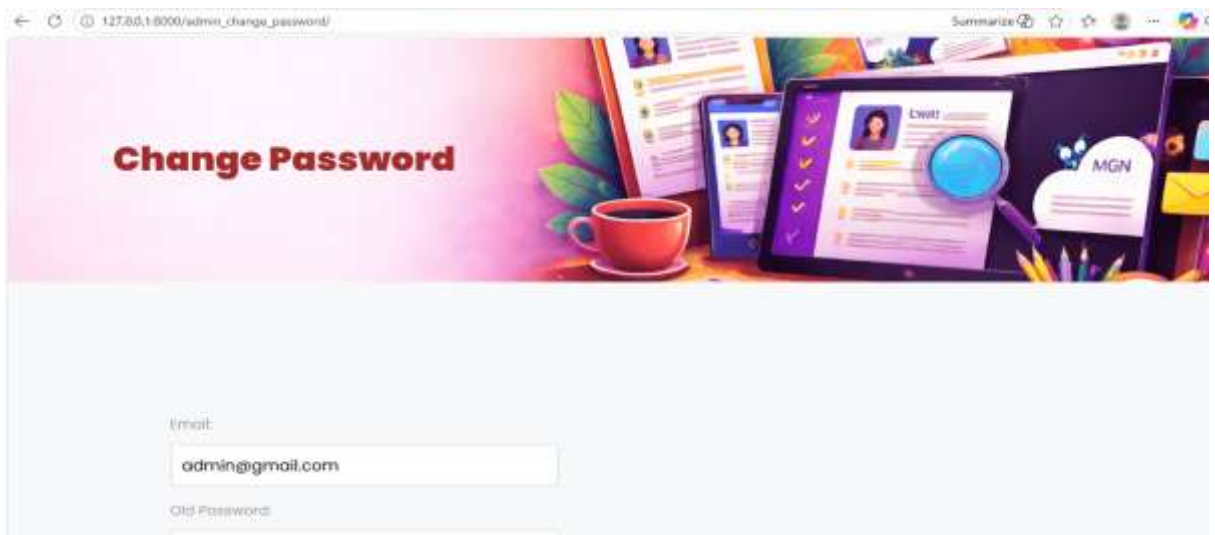
## Admin View Files:



## Admin View Users:



## Admin Change Password:



# TESTING

# Testing

## Testing Process:

### i. Introduction:

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

### ii. Types of Testing

#### Unit Testing:

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

#### Integration Testing:

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

#### System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

## User Acceptance Testing (UAT):

User Acceptance Testing (UAT), or application testing, is the final stage of any software development or change request lifecycle before go-live. UAT meaning the final stage of any development process to determine that the software does what it was designed to do in real-world situations. Actual users test the software to determine if it does what it was designed to do in real-world situations, validating changes made and assessing adherence to their organization's business requirements. The main purpose of acceptance testing is to validate end-to-end business flow.

User Acceptance Testing is a testing methodology where clients/end users participate in product testing to validate the product against their requirements. It is done at the client's site on the developer's site. For industries such as medicine or aerospace, contractual and regulatory compliance testing, and operational acceptance tests are also performed as part of user acceptance tests. It can be done in two levels i.e alpha testing and beta testing.



## Functional Testing:

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

## Non Functional Testing:

Non functional testing is a type of software testing that verifies non functional aspects of the product, such as performance, stability, and usability. Whereas functional testing verifies whether or not the product does what it is supposed to, non functional testing verifies how well the product performs.

### **White Box Testing:**

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

### **Black Box Testing:**

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

### **Test Strategy And Approach:**

Field testing will be performed manually and functional tests will be written in detail.

### **Test Objectives:**

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

### **Features To Be Tested:**

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

## i. Test Cases:

| <b>Id</b> | <b>Objective</b>              | <b>Description</b>                                                                                              | <b>Expected Result</b>                                                                      |
|-----------|-------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| log001    | To test the URL               | 1) Enter URL<br>2) Open the webpage<br>3) Check for GUI                                                         | Login validation page should open                                                           |
| log002    | To test the GUI               | 1) open the webpage<br>2) check the GUI components                                                              | It should display webpage with label,email,password,submit...etc                            |
| log003    | To validate the Login page    | 1) Open the login page<br>2) keep empty the input elements<br>3) click on login button                          | It should display error message as "Please enter email and password".                       |
| log004    | To validate the login page    | 1) Open the login page<br>2) Enter valid credentials<br>3) click on login button                                | It should display message as "Login successfully".                                          |
| mv005     | To validate the mobile number | 1) Open the login page<br>2) Enter mobile with less than 10 digits<br>3) click on login button                  | It should display error message as "Please enter mobile number with 10 digits".             |
| mv006     | To validate the mobile number | 1) Open the login page<br>2) Enter mobile with greater than 10 digits<br>3) click on login button               | It should display error message as "Please enter mobile number with 10 digits".             |
| mv007     | To validate the mobile number | 1) Open the login page<br>2) Enter mobile number starts with 6,7,8,9 digits<br>3) click on login button         | It should sent OTP to the respective mobile number                                          |
| mv008     | To validate the mobile number | 1) Open the login page<br>2) Enter mobile number doesnot starts with 6,7,8,9 digits<br>3) click on login button | It should display error message as "Please enter mobile number strats with 6,7,8,9 digits". |
| mv009     | To validate the mobile number | 1) Open the login page<br>2) Enter mobile number with special characters<br>3) click on login button            | It should display error message as "Please enter mobile number with valid digits            |

**Test Results :** All the test cases mentioned above passed successfully. No defects encountered.

# SCOPE

## **Future Scope**

### **Scope:.**

As the technology evolves, the web development industry promises noteworthy progress. We will see the industry growing and expanding with the introduction of new tools and technologies. Here is what the future of our website development are Responsive and Adaptive Designs, Payment Gateway Integration, Email / SMS Gateway integration, Mobile OTP Integrations, and also we can enhance this web application to mobile users by developing same application into mobile technology like android, IOS. So that user can use same application features even in mobile devices also. And also we can add GPS or location based services. You can even enhance the application by providing ML features like recommendations, predictions and classifications etc. We can also upgrade the best search and filter mechanisms too for this website.

# **CONCLUSION**

## Conclusion:

The **Smart Resume Writer and Cover Letter Builder Using GenAI** provides an efficient and intelligent solution for creating professional resumes and cover letters. By using Generative Artificial Intelligence, the system simplifies the process of document creation and reduces the time and effort required by users.

The system allows administrators to manage users, templates, notifications, and job postings effectively, while customers can easily create, preview, download, and manage their resumes. The feature of automatic and manual resume generation provides flexibility to users based on their needs. Additionally, the skill-based job notification system helps users find relevant job opportunities more easily.

This project demonstrates how modern technology can improve traditional processes and enhance user experience. In the future, the system can be further improved with advanced AI features, better customization options, and integration with external job platforms, making it a more powerful and user-friendly career support system.

Overall, the project demonstrates how thoughtful application of software engineering principles and architectures can solve real-world problems and enhance community welfare through technology.

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